

Bilateral Telemanipulation under Very Long Delays

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Dedication

To my family.

Acknowledgements

First and foremost I would like to express my gratitude to my supervisor Prof. Kostas Vlachos. As an advisor he always made time, was very patient and his suggestions was a major aid in the creation of this thesis.

I am also grateful to the friends i made over the years that made my academic journey worthwhile.

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Abstract

In the last decade, robotic systems advanced drastically in the mechanical domain. This improvement made them capable of a wider range of tasks, some of which are in remote locations, distant from the operator that is controlling the robot. For this reason, the need for better controllers that compensate for the extreme delay present in the teleoperation, and the complications it presents rose. Additionally, it is common for these systems to require some sensory data to be transmitted back to the operator, to aid in the perception of the environment the robot is operating in. Force feedback has been chosen as our returning sensory data, due to its ability to improve telepresence and being one of the two most important senses, along with vision, in terms of task and location perception. Also force application is a key feature that is required in most robotic operations.

As part of this study, we present one of the well-known methods of force-feedback bilateral teleoperation (wave variables) and its way of managing the complications that comes with the existence of large delays in the communication medium. We implement the control scheme on a simulated environment in addition to a real robotic system composed of a three degrees of freedom haptic device, we experiment on it and then present the resulting movement. Finally, we modify some of the characteristics of the controller to better suit a wider range of applications.

Περίληψη

Την τελευταία δεκαετία, τα ρομποτικά συστήματα έχουν εμφανίσει ραγδαία ανάπτυξη όσον αναφορά το μηχανικό τους μέρος. Αυτή η ανάπτυξη τους επέτρεψε να χρησιμοποιηθούν σε ένα μεγαλύτερο εύρος αποστολών, κάποιες εκ των οποίων πραγματοποιούνται σε απομακρυσμένες περιοχές, μακριά από τον χειριστή που ελέγχει το ρομπότ. Για αυτό τον λόγο, η ανάγκη για καλύτερα συστήματα ελέγχου που μπορούν να διαχειριστούν τις μεγάλες καθυστερήσεις στην επικοινωνία, όπως και τα παράγωγά τους προβλήματα, αυξήθηκε. Επιπρόσθετα, είναι σύνηθες αυτού του είδους τα συστήματα να μεταφέρουν προς τον χειριστή αισθητήρια σήματα, με σκοπό να τον βοηθήσουν στην αντίληψη του χώρου όπου το ρομπότ εγχειρεί. Επιλέξαμε το αισθητήριο σήμα που θα χρησιμοποιήσουμε στην εφαρμογή μας να είναι τύπου δύναμης, καθώς αποτελεί μαζί με την όραση, μια από τις δύο πιο σημαντικές μας αισθήσεις, στον τομέα της αντίληψης του χώρου. Επίσης, η εφαρμογή δύναμης είναι καίριο χαρακτηριστικό ενός ρομποτικού συστήματος, αναγκαίο στις περισσότερες εφαρμογές σήμερα.

Ως μέρος αυτής της έρευνας, παρουσιάζουμε μια από τις πιο δημοφιλείς μεθόδους διμερή τηλεχειρισμού με ανατροφοδότηση δύναμης (force-feedback bilateral teleoperation), όπως και τους τρόπους με τους οποίους η μέθοδος διαχειρίζεται τις μεγάλες καθυστερήσεις που υπάρχουν στην επικοινωνία μεταξύ του ρομπότ και του χειριστή. Εξετάσαμε το σύστημα ελέγχου αρχικά σε περιβάλλον προσομοίωσης και στην συνέχεια προσθέσαμε έναν αληθινό απτικό ρομποτικό βραχίονα τριών βαθμών ελευθερίας με την ικανότητα ανατροφοδότησης δύναμης στον χειριστή. Πειραματιστήκαμε με τα διαφορετικά περιβάλλοντα και παρουσιάζουμε τα αποτελέσματα των κινήσεων σε μορφή γραφημάτων. Τέλος, τροποποιήσαμε την συμπεριφορά του υποσυστήματος του χειριστή, προκειμένου να διευρύνουμε τις εφαρμογές του.

List of Figures

3.1	Normal controller without the use of waves or delay.	13
3.2	Normal controller without the use of waves but with delay.	14
3.3	2-Port network model of the teleoperation system.	15
3.4	Basic Master-Slave control scheme.	15
3.5	Position Tracking with adjusting the wave command scheme.	17
3.6	The Impedance matching approach control scheme.	19
3.7	Original impedance matching slave subsystem block diagram.	20
3.8	New slave impedance matching block diagram after one simplification. . . .	21
3.9	The final simplified block diagram of slave only impedance matching controller.	22
3.10	Final complete simplified block diagram of the impedance matching method.	23
4.1	The Novint Falcon haptic controller.	25
4.2	The complete teleoperation control scheme.	25
4.3	Main DLR space telerobotics missions communication parameters. P.Loss:Packet Loss; RT:Round Trip (Artigas and Hirzinger 2016).	26
4.4	Experiment with the simulated basic control scheme in free space	27
4.5	Experiment with the simulated online tuning control scheme in free space. .	28
4.6	Experiment with the simulated "adjusting the wave command" control scheme in free space.	28
4.7	Experiment with all three simulated subsystems combined (impedance matching, position tracking, online tuning) in free space.	29
4.8	Experiment with the simulated basic control scheme in free space with an immovable object.	30
4.9	Experiment with the simulated online tuning control scheme in free space with an immovable object.	30
4.10	Experiment with the simulated "adjusting the wave command" control scheme in free space with an immovable object.	31
4.11	Experiment with all three simulated subsystems combined (impedance matching, position tracking, online tuning) in free space with an immovable object.	32
4.12	Experiment with the simulated basic control scheme in free space with an immovable object and friction.	33
4.13	Experiment with the simulated "adjusting the wave command" control scheme in free space with an immovable object and friction.	33
4.14	Experiment with the simulated "adjusting the wave command" control scheme in free space with an immovable object and friction.	34
4.15	Experiment with all three simulated subsystems combined (impedance matching, position tracking, online tuning) in free space with an immovable object and friction.	35
4.16	Experiment with the basic control scheme and Novint Falcon in free space.	36

4.17 Experiment with the online tuning control scheme and Novint Falcon in free space.	36
4.18 Experiment with the "adjusting the wave command" control scheme and Novint Falcon in free space.	37
4.19 Experiment with all three subsystems combined (impedance matching, position tracking, online tuning) and Novint Falcon in free space.	38
4.20 Experiment with the basic control scheme and Novint Falcon in free space with an immovable object.	38
4.21 Experiment with the online tuning control scheme and Novint Falcon in free space with an immovable object.	39
4.22 Position decoupling in experiment with the online tuning control scheme and Novint Falcon in free space with an immovable object.	40
4.23 Experiment with all three subsystems combined (impedance matching, position tracking, online tuning) and Novint Falcon in free space with an immovable object.	41
4.24 Experiment with the basic control scheme and Novint Falcon in free space with added friction.	42
4.25 Experiment with the online tuning control scheme and Novint Falcon in free space with added friction.	42
4.26 Experiment with the position tracking subsystem and Novint Falcon in free space with added friction.	43
4.27 Experiment with all three subsystems combined (impedance matching, position tracking, online tuning) and Novint Falcon in free space with added friction.	44
4.28 Experiment with the basic control scheme and Novint Falcon in free space with an immovable object and added friction.	44
4.29 Experiment with the online tuning control scheme and Novint Falcon in free space with an immovable object and added friction.	45
4.30 Experiment with all three subsystems combined (impedance matching, position tracking, online tuning) and Novint Falcon in free space with an immovable object and added friction.	46
4.31 Experiment with two subsystems combined (impedance matching, online tuning) and Novint Falcon in free space with an immovable object.	47
4.32 Velocity control block diagram.	47

Table of Contents

1	Introduction	8
1.1	Wave variables	8
1.2	Thesis Structure	8
2	Review of the Literature	10
3	The wave variables approach	13
3.1	The basic wave model	14
3.2	Position drift	16
3.2.1	Transmitting the Wave Integrals	16
3.2.2	Adjusting the Wave Command	16
3.3	Wave reflections	17
3.3.1	Wave Filtering	18
3.3.2	Impedance matching	18
4	Experiments	24
4.1	The system	24
4.2	Modeling and Simulation	26
4.2.1	Simulation in free-space	27
4.2.2	Simulation in free-space with an obstacle	29
4.2.3	Improved master and slave models	32
4.3	Experimental Implementation	35
4.3.1	Novint Falcon	35
4.3.2	Improved Slave Model	40
4.3.3	Velocity Control	46
5	Conclusions and Future Work	49

Chapter 1

Introduction

During the bilateral teleoperation of a robotic system over large communication delays, a common issue that occurs is that the feedback information, which in this case is in the form of force from the robot, affects the haptic device of the operator in such a way, that combined with the operator's input, makes the system unstable. The result is sharp movements and a big escalation of position and velocity, both in the robot and the haptic device. Ideally the operator wants to feel through the haptic device the forces the robot is facing, by pushing an object or in extreme cases, colliding with an immovable wall. Researchers found out that if you can ensure the passivity of the system, then it will also be stable for as long as the passivity criteria are met. There are a few methods that try to tackle this problem in many different ways, but a prominent one is the wave variable approach.

1.1 Wave variables

Introduced in 1991 (Gunter Niemeyer and Slotine [1991](#)), the wave variable approach addressed the issue of passivity by combining the force and velocity commands into a single variable. The transformation is bijective, so that it is always unique and invertible and only the wave variable needs to be transmitted to the robotic device and back to the operator. This means that the wave itself does not distinguish between velocity or force and the recipient cannot detect the original form. Instead, any element may interpret an incoming wave as best suited to its current needs, so a robot in contact with the environment will use an incoming wave command to generate force, whereas a robot in free space will use the same wave command to generate motion. To satisfy passivity, a wave response is limited in magnitude to the original wave command and its stored energy.

The approach comes with two main issues, position drift and wave reflections. Position drift is when the position of the robot starts to deviate from the position of the master, considering a slave-master system, due to discrete sampling of the velocities, numerical errors from finite precision and any data losses that happen as a consequence of the transmission. The ways these problems are solved, will be presented and discussed in the following chapters.

1.2 Thesis Structure

The remainder of this Thesis will be structured as follows. Chapter 2 presents a quick overview of the related literature that was used in the creation of this thesis. Chapter 3 covers the basic information on the wave variables method, needed for the rest of the paper, the theory of the two major problems that are inherent to the approach and four distinct ways of dealing with said problems. It also explains the Online Tuning process and its uses,

while presenting a control scheme for every method that will be used in the experiments chapter. Finally, an impedance matching block diagram simplification is constructed along its mathematical analysis. Chapter 4 contains the experiments conducted on three different subsystems and their combination, in two different environments. A simulation and an improved simulated slave model will be used, along with a real life haptic device as the master robot and in the final parts we will change the behavior of the master subsystem to a velocity controller to better match a wider range of applications. Chapter 5 concludes our work, discuss the results of the experiments and proposes future avenues for the contents of this thesis.

Chapter 2

Review of the Literature

As described in the introductory parts of this paper, the need for robotic teleoperation through large delays caused either by distance or by a poor medium of communication, is only increasing. Another desired feature of these robotic systems is the ability to send back to the operator sensory data, like force, which then a haptic device can help the operator feel what the robotic system is "feeling". Here we will focus only on force feedback as the sensory data that will be transmitted back to the operator.

Even though communication with a robotic system through large delays introduces many problems to the system, adding force feedback capabilities and changing the communication to bilateral to support this need, bring in another big set of problems that needs to be addressed. The most prominent of these problems is stability, which will be analysed in depth in the next chapters.

One of the main approaches to solving this problem is through the use of the wave variables, introduced by Günter Niemeyer and Jean-Jacques E. Slotine in 1991 in their paper (Gunter Niemeyer and Slotine 1991). Through their future work (Gunter Niemeyer and Slotine 1997), (Günter Niemeyer and Slotine 1998) and (Günter Niemeyer and Slotine 2004) 3 main problems surface, passivity, position drift and wave reflections, the last two of which arise from the wave variables approach. For dealing with the inherent problem of wave reflections they suggest two different ways, impedance matching and wave filtering.

The wave filtering is applied by using two first order linear filters, one in each transmission direction. They point out that with this approach is analogous to directly damping the wave communications, instead of relying on the dissipation in the master/slave subsystems to absorb wave reflections and oscillations.

The other approach they analyse to reduce the effects of wave reflections is to use the impedance matching method. This is implemented by introducing two dissipation elements in both impedance controllers, R and D. D is used to create a critically damped response, necessary for the dynamic simplification. The damping R is added to match the plain controller impedance to the wave impedance and help dissipate the communications resonance if needed. This method is better than the wave filtering method and should be proffered if and when is possible.

Going back to the 3 main problems wave variables deal with, position drift is also addressed by G.Niemeyer and J.Slotine in two different ways. Adjusting the wave command is their first method, in which the goal is to modify the wave command with a corrective term computed from the difference of the master and slave positions. This means that we need to add another signal to our transmission from the robot back to the operator, the robots position. We will go into further details at the next chapter where we will use this method to negate the inherit drift that occurs from using the wave variables method.

The second method they develop to counteract the position drift is by transmitting the wave integrals. Because the wave signal encodes the velocities and forces of the master-

slave system, the integral of this wave signal encodes position and momentum information. So by transmitting both wave and integrated wave signal the system can bind the two positions preventing any drift. This method comes with the benefit of not needing an additional transmission channel. We can combine the two signals and transmit the simultaneously. This is possible due to the fact that the two values are closely related via integration or differentiation.

Passivity, we can say, is the most important of the 3 phenomena, and is one of the reason the wave variables method was invented. To ensure passivity, a wave response is limited in magnitude to the original wave command. This condition states that more energy must be introduced into a passive system than can be removed from it, subject to the initial stored energy. Also, by applying extra systems to mitigate the wave reflections and the position drift, we must test if the passivity arguments are violated. In their papers G.Niemeyer and J.Slotine ensure that all 4 methods (adjusting the wave command, transmitting the wave integral, impedance matching and wave filtering) are created as such to comply to the passivity arguments.

Thinking of an appropriate transmission framework, an internet based teleoperation comes naturally in mind, due to its extremely wide use in long distance communications, its wide spread infrastructure and the low cost for using it. In their paper, Roberto Oboe and Paolo Fiorin (Oboe and Fiorini 1997) describe the some of the issues relevant to the use of Internet within a control system, and report on experiments of Internet modeling and of Internet-based teleoperation. Two inherent problems that come with the use of internet within a control system are delay jitter and data losses. They proceed to demonstrate these features in a few communication experiments, and show the range of values the delay and data losses take in the year 1997. They observe that despite its shortcomings, in a non-congested network, internet hold interesting promises, while the design of such suitable controller for a system that integrates an Internet connection, is an open research field.

The next year Kevin Brady and Tzyh-Jong Tarn, in their work (Kevin Brady and Tzyh Jong Tarn 1998) presented a promising architecture of a controller in the field of internet-based control. They use a predictive robotic model with they complement with a live delayed video feed from the remote robotic system. They also point out the random, time-varying communication propagation delay problems between the teleoperator, the important role of bandwidth and its relationship with propagation delay, as well as their limitations. Examples of different internet-based teleoperation implementations are shown in addition to the state space model they use. Finally, from their experimental results they conclude that the safety, efficiency, telepresence and transparency goals of the internet-based "human in the loop" architecture proposed, have been met. Lastly in their future work (K. Brady and T. J. Tarn 2000) Kevin Brady and Tzyh-Jong Tarn observe that internet-based teleoperation has moved into the workplace in the early 2000s.

We continue on with a few internet-based, long distance robotic teleoperation applications in space, which was the inspiration for this thesis on the subject of long distance, force-feedback teleoperation. In (Backes, Tso, and Tharp 1998), Paul G. Backes, Kam S. Tso and Gregory K. Tharp introduce (WITS) the Web Interface for telescience which was an Internet-based tool that the Mars Pathfinder mission used for both mission operations and will also be used in the 1998 lander and 2001, 2009, and 2005 rover missions to Mars. WITS consists of an client and a server part. The server part is resides and executes at JPL, while the client part can be downloaded and executed on any user's computer. Communication with the rover is made through the server part of WITS with radio waves, but the client part allow any scientist communicate, from their home lab, with JPL and in extend the rover. This is possible because WITS uses the internet for its client communications, and is the main reason this system is implemented.

To get a sense of the environment and complete system our analysed control architecture will be inspired to work, we study two great experiments that happened in collaboration with the International Space Station, DLR's ROKVISS and KONTUR-2. In their papers (Albu-Schäffer et al. 2006), (Preusche et al. 2006), (Landzettel et al. 2006), (Artigas and Hirzinger 2016) and (Artigas, Balachandran, et al. 2016) the complete mission system, a brief history of DLR's space telerobotics and force-feedback teleoperation, and also five different space teleoperation programmes (ROKVISS, KONTUR-2, TECSAS, CESOR and Rotex the first space telerobotics mission in history) are analysed. We see the different communication infrastructure they used as well as the impact it had on the communication parameters, the range delay, jitter, bandwidth and other important metrics take, the distinct control methods and the way they tackle the passivity problem, and most important, the standards a force-feedback teleoperation through large delays system must comply with to be useful in these applications.

Another very popular approach to teleoperation with time delays is the time domain passivity control method. This approach is demonstrated by Jordi Artigas, Jordi Vilanova, Carsten Preusche and Gerd Hirzinger in their work (Artigas, Vilanova, et al. 2006). This technique despite its effectiveness will not be used in our control scheme, so we will not analyse it deeply.

At last Da Sun, Fazel Naghdy and Haiping Du in (Sun, Naghdy, and Du 2014) conducted a complete survey on the research space of wave variable control for bilateral teleoperation systems. They evaluated the different ways to mitigate the inherent problems of the wave variable method like the ones we introduces earlier, a few techniques developed to improve on the idea have been assessed, and finally the research gaps in this field are identified and future directions for further research are proposed. This is a very important paper for us, as it demonstrates the ways the wave variable approach have evolved and improved, as well as a very explanatory way in all the details of this method.

Chapter 3

The wave variables approach

Before we introduce the wave variables method of dealing with passivity, let's first examine the reaction to delay of a normal master-slave controller without the use of waves. The system will be simulated and comprised of the basic mass models of the master and slave robots and a PD controller added to the slave subsystem to match the desired velocity to the actual velocity of the slave. As we will see in the next section, this system is similar to the basic wave model besides the use of any waves.

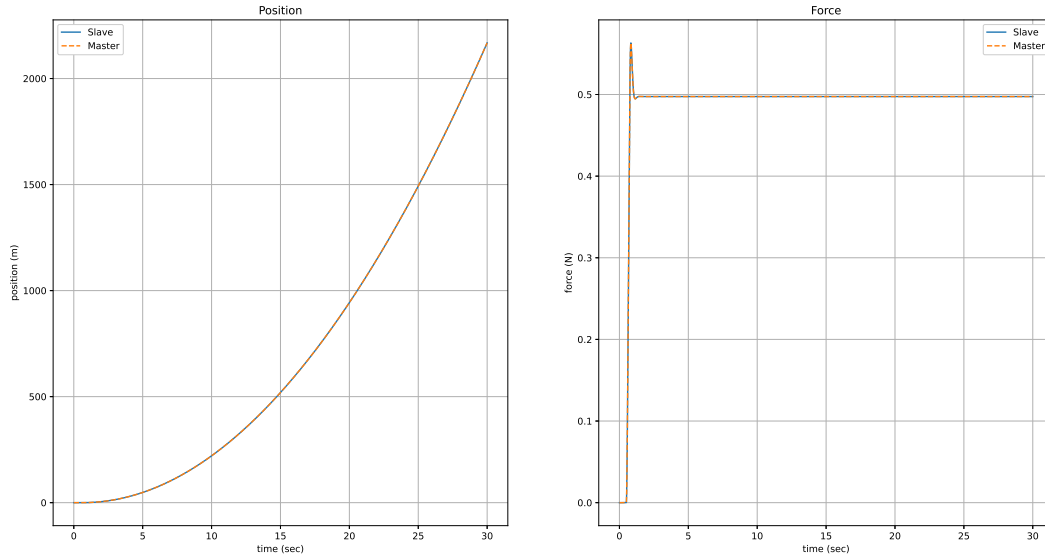


Figure 3.1: Normal controller without the use of waves or delay.

We can observe that without the use of delay the motion of both master and slave robots is normal and by using a constant force as input of the master, the two positions increase steadily. When we add delay to the system, firstly we see an initial divergence of the two positions, caused by the delayed reaction of slave. But more importantly after a few seconds, oscillations start to appear in the forces and positions of both subsystems and eventually lead them into instability.

This exact phenomenon we will try to counteract with the use of wave variables in the next section of this Thesis.

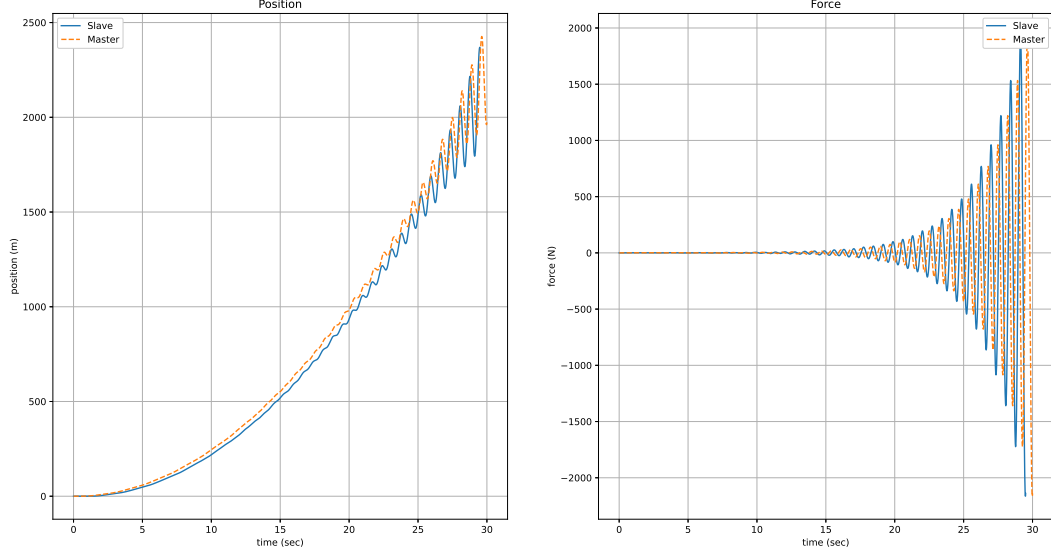


Figure 3.2: Normal controller without the use of waves but with delay.

3.1 The basic wave model

As stated in the introductory parts of this study, the wave variables method uses a transformation of the velocities and forces present in the master or slave half of the system, which then are transmitted to the rest of the system. We can distinguish them into a forward or right moving wave denoted as u and a backwards or left moving one, denoted as v . The transformations that compute these values are:

$$u = \frac{b\dot{x} + F}{\sqrt{2b}} \quad v = \frac{b\dot{x} - F}{\sqrt{2b}}$$

The specific force F and the specific velocity $\dot{x}$ will change depending on the model we are using. The wave impedance denoted as b is a positive constant that assumes the role of a tuning parameter trading off speed of motion against level of forces. Inverting these transformations allow the opposite side of the transmission to decode the information required for the task. In practise tuning the parameter b will change the characteristics of the command transmitted from the operator to the slave. Higher b values allow the correct transmission of forces, minimising the movement of the robot. This is useful when the robotic system is tasked to push an obstacle or an immovable object, allowing the system to transmit the correct forces from the operator, through the robot and to the obstacle, as well as transmitting the exact feedback forces the robot is facing, back to the hands of the operator. Lower values of b , prioritize the movement commands and allowing the robotic system to match the velocity and consequently the position of the master. This behavior is of use in tasks that include maneuvering in free space while matching the velocity of the operator without pushing of obstacles.

Rarely though a real task consists only of movement or push commands. For that reason we need to tune the wave impedance accordingly, and on-line, dependent on the current task. High wave impedance values enables sensitive force feedback to take place, whereas low values allows fast motions to be executed. Also, by increasing b right before a contact with the environment, we prevent big contact force spikes to appear, due to the initial hit. Hence the impedance matched teleoperator, together with appropriate on-line tuning of the wave impedance, can achieve good performance both in free space and in contact.

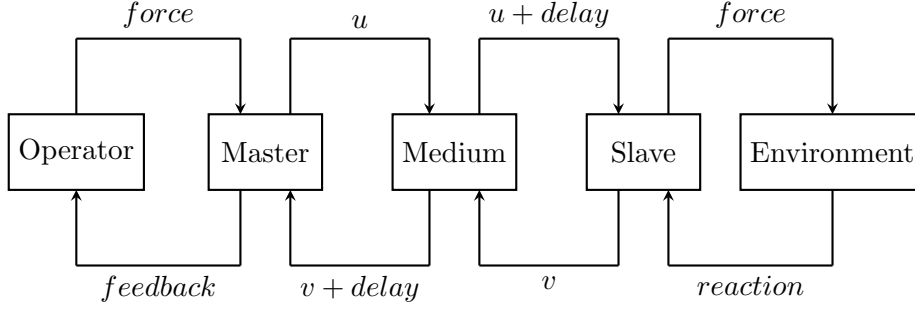


Figure 3.3: 2-Port network model of the teleoperation system.

To complete this basic wave teleoperator, we will place a PD controller in the slave robot system to try and match the desired velocity that is decoded from the right moving wave, to the actual velocity the slave robot sensors measure. To remind how a PID controller functions we need to brake it into three parts, the proportional, the integral and the derivative. The proportional section of the controller, responsible for the "P" part in the name, is correcting the controlled value by adding a term proportional to the difference of the control and the controlled value, in our case $\dot{x}_{sd}$ and $\dot{x}$ respectively. Similarly, the integral part is responsible for the "I" in the name of the controller, and it works by using the past information of the movement, adding a value proportional to the duration and magnitude of the error in an attempt to correct the steady state error created by the proportional part of the controller. Finally, the derivative section creates an output signal proportional to the rate of change of the error, practically trying to see into the future of the movement and contribute accordingly.

Using these systems, we arrive at the following control scheme, where the master and slave robots are represented as simple masses.

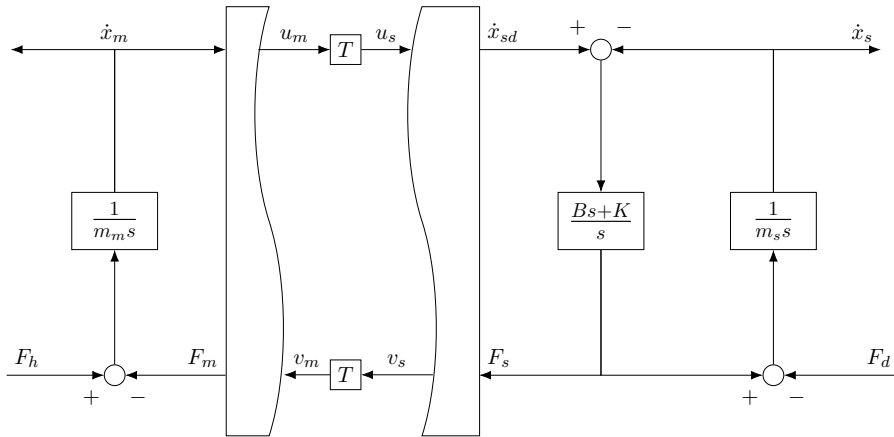


Figure 3.4: Basic Master-Slave control scheme.

Talking about passivity, it is the main reason the wave variables were initially conceived. The passivity condition states that more energy must be introduced into a passive system than can be removed from it, considering the initial stored energy. In wave variable terms this means that the outgoing wave (u_m or v_s) is limited by the energy carried by the incoming wave (u_s or v_m) plus the initial stored energy. If we think of time delay in the wave domain as a temporary storage while the waves are in transit, then the power dissipation of the input wave is zero, making the communications not only passive, but also lossless. All the subsystems that we will introduce in the following sections also preserve the passivity criteria as the basic wave mechanic remains unaltered.

Even though this basic system is capable of dealing with the problem of large delay instability, it lacks the systems that will prevent the issues that are inherent to the wave variables method from appearing. The most prominent are position drift and wave reflections.

3.2 Position drift

In the beginning of this paper, we discussed the meaning of the term "position drift" and the reasons of its occurrence. Many attempts have been made with grate success, like in their work (L. Y. Hu, X. P. Liu, and G. P. Liu 2010) and (L. Hu et al. 2011). Here we will focus on the efforts of the authors G.Niemeyer, J.Slotine and Jean Jacques E. and their two distinct solutions to the problem, we will link one of these approaches to the complete system and we will test its effectiveness.

3.2.1 Transmitting the Wave Integrals

While the previous basic model of the wave-based teleoperation will force the master-slave velocities to match each other, the essence of the following two methods is to couple the two positions by adding the position information to the communication.

The first method "Transmitting the wave integrals" requires two concepts to be explained, the meaning of a wave integral and the way the added information will join the waves in its transmission.

The new integrated wave variables encode position and momentum information instead of the robots velocity and force. In essence we transmit the master position by encoding it along with previous left moving wave integral signal. On the slave side, the desired position can be decoded by using the right moving integrated wave with the momentum information, which we get by integrating the force slave applies into the environment. These developments do not effect the passivity criteria and can be incorporated freely into any wave-based teleoperation system (Günter Niemeyer and Slotine 2004).

Now, transmitting this new value might seem to require a new channel in the communication system. This can be overcome by combining the two signals into one, the wave integral as it is and the normal wave adjusted by a value. On the opposite side of the system, the information can be decoded by a stable first-order filter and used as normal. So, through this system we managed to include all the required information for a stable position tracked wave-based teleoperation into a single value.

3.2.2 Adjusting the Wave Command

The wave-based position tracking approach we chose to implement is the "Adjusting the Wave command" approach. The main goal of this, as the previous one, is to use the slave position to slowly match with the master position. It accomplices that by adding a corrective term into the transmission of the basic wave signal. This terms job is to correct for the position drift that occurs between the master-slave system. The added system also has the ability to correct for any initial position offsets and temporary transmission losses without introducing any sudden step inputs.

To do this, as we seen in Figure 3.5, we need to add another backward communication path that is used to transmit the slave position back to the master. There, a predicted position difference is calculated based on previous information, after which in combination with the actual position difference, calculated from the two master-slave positions, a drift error term is created. Hence, the corrective term is:

$$\Delta u_m = -\sqrt{2b\lambda}d$$

where $\mathbf{b}$ is the standard wave impedance, $\mathbf{d}$ is the previously calculated drift error and λ is the bandwidth of this new feedback path. The drift error $\mathbf{d}$ and the corrected wave term $\Delta \mathbf{u}_m$ must always be of the opposite sign, so the path's bandwidth must be selected appropriately.

In addition, we need to modify the wave correction term based on each degree of freedom. The resulting $\Delta \mathbf{u}_m$ term and the drift error is never of the same sign assuring that $\mathbf{d}$ will never increase. A problem of this system though is that it displays a "waiting"

$$\Delta u_m = \begin{cases} 0 & \text{if } d\hat{u}_m < 0 \\ -\sqrt{2b\lambda}d & \text{if } d\hat{u}_m > 0 \text{ and } \sqrt{2b\lambda}|d| < |\hat{u}_m| \\ -\hat{u}_m & \text{if } d\hat{u}_m > 0 \text{ and } \sqrt{2b\lambda}|d| > |\hat{u}_m| \end{cases} \quad (3.1)$$

behavior as it requires energy from the operator in order to force position tracking. This means that in situations that the operator is inactive, or offline, the system retains its position discrepancy between master and slave.

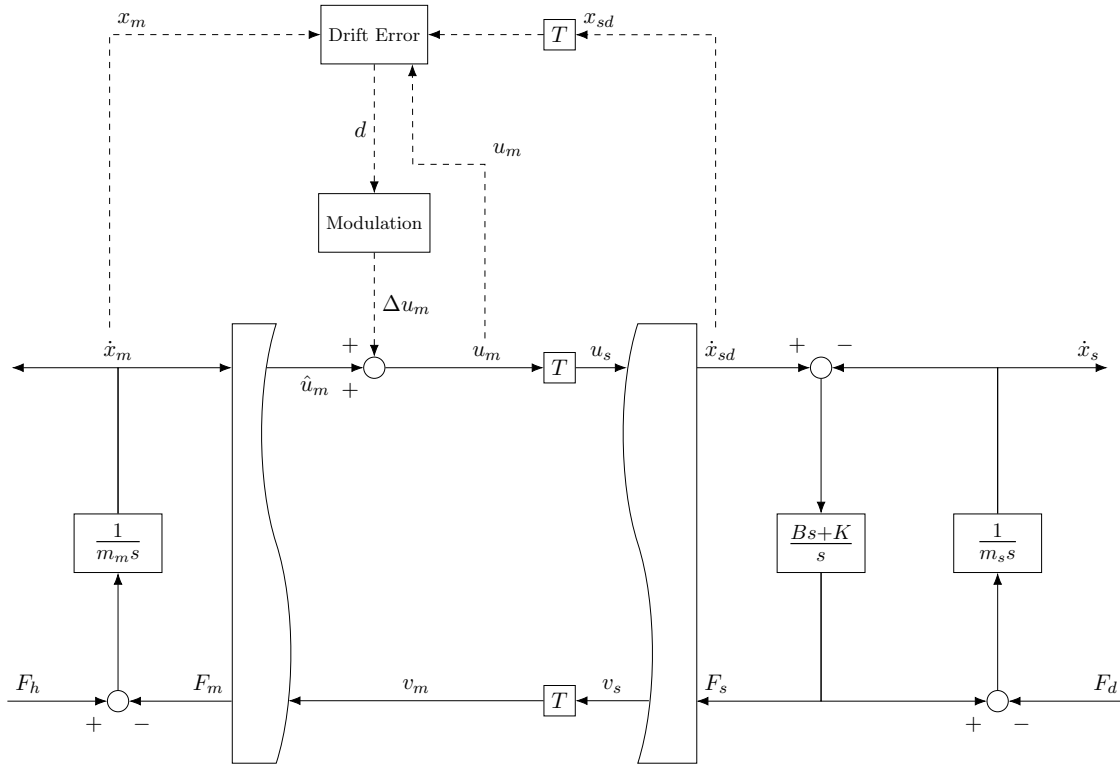


Figure 3.5: Position Tracking with adjusting the wave command scheme.

In regards of passivity, the new corrective's term's energy must be bound by the normal wave because it does not contain any power. In effect to preserve the passivity criteria the new feedback path only uses "left over" power, not used by the main system. So, having satisfied passivity, we see that the new system can be used in any wave-based teleoperation system if tuned accordingly.

3.3 Wave reflections

The second major drawback of a wave-based teleoperation is the apparent "wave reflections" caused by one of the feedback loops existing in the basic system. This feedback loop appears since the information encoded and transmitted back to the master system from slave, is dependant on the signal it received exactly prior to the coding operation

from the environment, but most importantly, from master. This behavior is repeated in the master side as it needs the information slave transmitted T second prior, to encode the next signal the human operator commanded. In this procedure, there is residual information from past signals, maintained for a few cycles, altering the commands and changing the systems behavior. Maybe, under extreme circumstances and delay, leading it into instability. In other words, Wave reflections is a phenomenon caused by the oscillation in the response of the teleoperation system.

To counteract this residual information, in their work (Gunter Niemeyer and Slotine 1991),(Günter Niemeyer 1996)(Gunter Niemeyer and Slotine 1997),(Günter Niemeyer and Slotine 1998) and (Günter Niemeyer and Slotine 2004) presented two prominent methods, impedance matching and wave filtering.

3.3.1 Wave Filtering

The wave filtering method should be used if the impedance matching approach, discussed in the next section of this thesis, is unable to produce the desired results in the system, or can not be implemented entirely. The reason for this is that by filtering the wave signals, as the name implies, to reduce the effects of any reflections enduring in the signals circulating the controllers, we greatly decrease tracking performance by smoothing out its behaviour regardless of the task. Specifically, high frequency information transmitted by the operator or the environment are similar in nature with the wave reflections, and for that reason they get filtered out easily by the filter.

The chosen filter can be of any type as long as its gain is limited below unity and is able to filter out the high frequencies appearing through wave reflections. Even something as simple as a first-order linear low pass filter used in a few papers discussing the subject. As for passivity, adding a filter into the wave transmission path does not affect the criteria, so the system retains its passivity in any environment.

3.3.2 Impedance matching

Wave reflections appear when a wave signal hits an element with an impedance different to its own wave impedance b . Impedance matching is our method of choice for negating wave reflections. Derived from the aforementioned papers, it is the prevailing method of the two, based on the lesser drawbacks and its effectiveness. The main idea is to make the master and slave subsystem both appear like a simple damper to the wave signals, this way the incoming energy is dissipated and no reflections can occur. To achieve this, we match the impedance of the master and slave subsystems by creating two symmetric controllers.

Two damping elements $\mathbf{R}$ and $\mathbf{D}$ are introduced at both sides. Combined with the PD controller they constitute an impedance controller with parameters:

$$D = \lambda m \quad B = \lambda m \quad K = \lambda^2 m \quad R = b - \lambda m$$

where the λ term is referred to the bandwidth of the system and can be tuned independently to best match the characteristics of each. The damping $\mathbf{D}$ is used to create a critically damped response to the system, while the damping $\mathbf{R}$ is added to match the impedance of the wave to the controller impedance. These values for the damping terms have been selected because by tuning the controller this way we achieve to simplify the dynamics of the model to a simple critically damped PD controller.

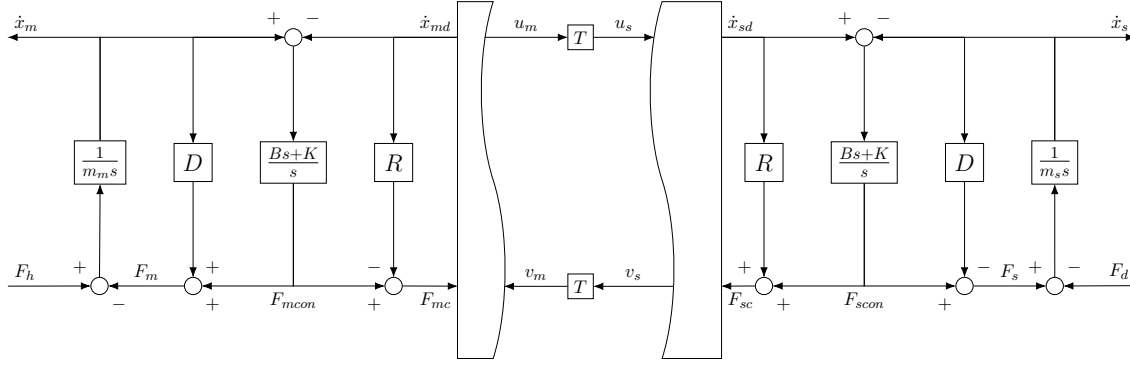


Figure 3.6: The Impedance matching approach control scheme.

The controller can be split into two parts for each master and slave subsystem, the slave or master robot model and the rest of the controller. The outputs of the controller part are the controller force, that gets transformed into the wave response, and the F_s/F_m signal which is the force the controller is applying into the slave or master robot. These forces are calculated from the following equations.

$$F_m = F_{mcon} + \dot{x}_m D = \dot{x}_m D + B(\dot{x}_m - \dot{x}_{md}) + K(x_m - x_{md})$$

$$F_s = F_{scon} - \dot{x}_s D = -\dot{x}_s D + B(\dot{x}_{sd} - \dot{x}_s) + K(x_{sd} - x_s)$$

A tuning operation is also needed for the system to display the behavior we require for each task. The wave impedance must be selected base on the task the robot is operation in, just like the on-line tuning procedure we described earlier. For mainly freespace movements we must select low values for the impedance, allowing the minimum stiffness in the system and great agility. Larger impedance values are needed for contact "push" tasks, where the contact forces from slave, as well as the applied forces form the human operator, need to be transmitted as accurately as possible. For hybrid movements, an on-line tuning system must also be implemented.

Though, before tuning the wave impedance for the current environment, the systems bandwidth λ must be firstly tuned, both in the slave subsystem and the master subsystem. This is important to ensure that they both appear as a pure damper to the wave commands and necessary to further tune the system.

Noteworthy is the fact that, by tuning the wave impedance as low as possible, we essentially remove the force feedback capabilities of the system and transform it into a pure movement controller. On the contrary, very high wave impedance values restricts completely the robots ability to move and allow it to only transmit forces from the operator. The same behavior exhibits the operators haptic device, as the returning forces, even without contact, are amplified, preventing movement even in freespace. In essence, the wave impedance assumes the role of a rough predictor of the task and environment the system is operating in.

Lastly, to ensure impedance matching the wave impedance must always be greater than or equal to the master or slave dissipation, because we need the controller to be passive and it can only apply damping to the system. For this reason we have the following restriction:

$$b \geq B = D = \lambda m$$

Impedance matching block diagram simplification

After analysing the impedance matching method to prevent wave reflections of appearing and before we construct the system, it is interesting to develop the simplified block

diagram which uses only four transfer functions on each side. Four transfer functions is the minimum possible number slave can have, as it utilises two inputs and two outputs, and we need a function to describe the behavior of one output based on the first input, and a second function for the other input to the system. Master behaves the same way, since it also has a pair of inputs and a pair of outputs. To start, we will focus on the slave subsystem, but as the system is symmetrical, only a few changes in the equations will be needed to extrapolate the master side.

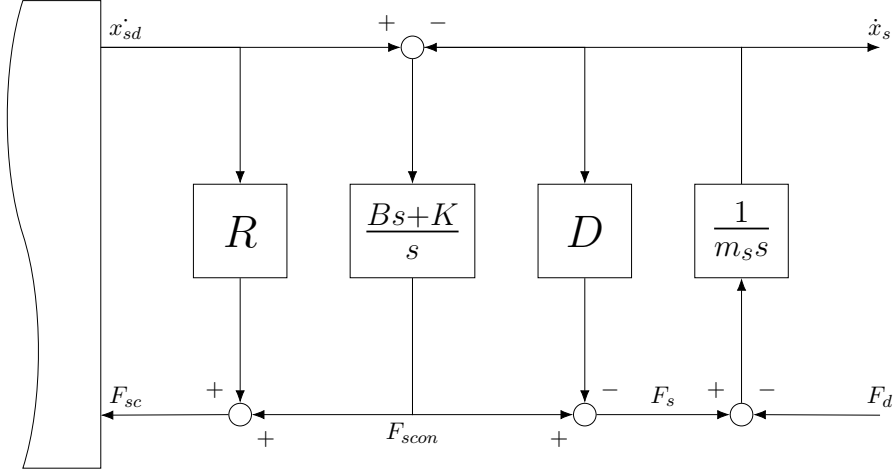


Figure 3.7: Original impedance matching slave subsystem block diagram.

We begin with the slave half of the original block diagram from which we get the following equations in the Laplace domain. For the sake of not repeating the Laplace transformation symbol $\mathcal{L}$ in the following equations, we assume that $\dot{x}_s = \mathcal{L}\{\dot{x}_s\}$, $\dot{x}_{sd} = \mathcal{L}\{\dot{x}_{sd}\}$, $\dot{F}_s = \mathcal{L}\{\dot{F}_s\}$, $\dot{F}_{sc} = \mathcal{L}\{\dot{F}_{sc}\}$ and $\dot{F}_d = \mathcal{L}\{\dot{F}_d\}$ **only** symbolically. So, we have for the output $\dot{x}_s$:

$$\dot{x}_s = (F_s - F_d) \frac{1}{m_s s} = \frac{F_s}{m_s s} = \frac{-\dot{x}_s D + F_{scon}}{m_s s} \Rightarrow \dot{x}_s \left(1 + \frac{D}{m_s s}\right) = \frac{F_{scon}}{m_s s} \Rightarrow \frac{\dot{x}_s}{F_{scon}} = \frac{1}{m_s s + D} \quad (1)$$

$$\begin{aligned} \dot{x}_s &= (F_s - F_d) \frac{1}{m_s s} = \frac{F_s}{m_s s} - \frac{F_d}{m_s s} = \frac{-\dot{x}_s D + F_{scon}}{m_s s} - \frac{F_d}{m_s s} = \frac{-\dot{x}_s D}{m_s s} - \frac{F_d}{m_s s} \Rightarrow \\ &\Rightarrow \dot{x}_s \left(1 + \frac{D}{m_s s}\right) = -\frac{F_d}{m_s s} \Rightarrow \frac{\dot{x}_s}{F_d} = -\frac{1}{m_s s + D} \quad (2) \end{aligned}$$

From (1) and (2):

$$\dot{x}_s = \frac{F_{scon}}{m_s s + D} - \frac{F_d}{m_s s + D} = \frac{1}{m_s s + D} (F_{scon} - F_d) \quad (3)$$

The resulting block diagram is:

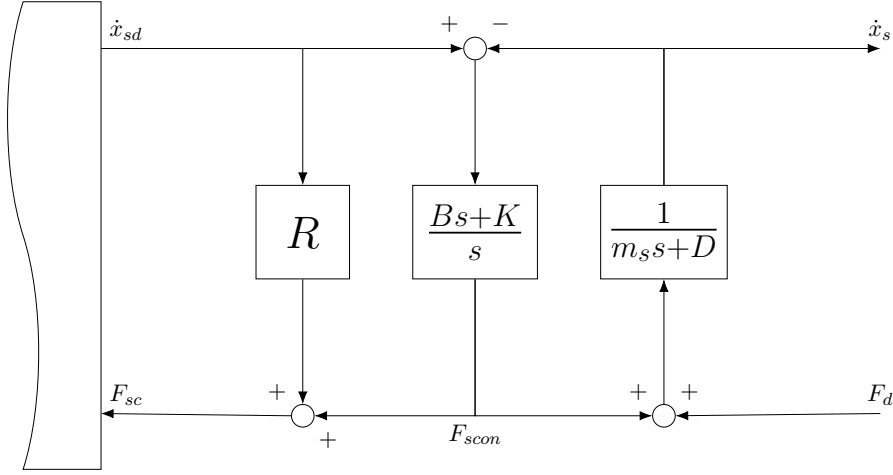


Figure 3.8: New slave impedance matching block diagram after one simplification.

Next, we try to simplify the the block diagram even further while keeping our focus at the transfer functions of $\dot{x}_s$.

$$\begin{aligned}\dot{x}_s &= \frac{F_{scon}}{m_s s + D} - \frac{F_d}{m_s s + D} = \frac{1}{m_s s + D} \frac{Bs + K}{s} (\dot{x}_{sd} - \dot{x}_s) = \frac{Bs + K}{m_s s^2 + Ds} (\dot{x}_{sd} - \dot{x}_s) \Rightarrow \\ \Rightarrow \dot{x}_s \left(1 + \frac{Bs + K}{m_s s^2 + Ds}\right) &= \dot{x}_{sd} \frac{Bs + K}{m_s s^2 + Ds} \Rightarrow \dot{x}_s = \dot{x}_{sd} \frac{(m_s s^2 + Ds)(Bs + K)}{(m_s s^2 + Ds)(m_s s + Ds + Bs + K)} \Rightarrow \\ &\Rightarrow \dot{x}_s = \dot{x}_{sd} \frac{Bs + K}{m_s s^2 + Ds + Bs + K} \quad (4)\end{aligned}$$

$$\begin{aligned}\dot{x}_s &= \frac{F_{scon}}{m_s s + D} - \frac{F_d}{m_s s + D} = \frac{1}{m_s s + D} \left[\frac{Bs + K}{s} (\dot{x}_{sd} - \dot{x}_s) - F_d \right] = \frac{1}{m_s s + D} \left[-\dot{x}_s \frac{Bs + K}{s} - F_d \right] \Rightarrow \\ \Rightarrow \dot{x}_s \left(1 + \frac{Bs + K}{m_s s^2 + Ds}\right) &= -F_d \frac{1}{m_s s + D} \Rightarrow \dot{x}_s = -F_d \frac{(m_s s + D)s}{(m_s s + D)(m_s s^2 + Bs + Ds + K)} \Rightarrow \\ &\Rightarrow \dot{x}_s = -F_d \frac{s}{m_s s^2 + Bs + Ds + K} \quad (5)\end{aligned}$$

Using (4) and (5) we derive the following transfer function.

$$\dot{x}_s = \dot{x}_{sd} \frac{Bs + K}{m_s s^2 + Ds + Bs + K} - F_d \frac{s}{m_s s^2 + Bs + Ds + K} \quad (6a)$$

OR

$$\dot{x}_s = \frac{s}{m_s s^2 + Bs + Ds + K} \left[\dot{x}_{sd} \frac{Bs + K}{s} - F_d \right] \quad (6b)$$

Lastly in our simplification process, we target the output F_{sc} and try to untangle its transfer function.

$$\begin{aligned}F_{sc} &= \dot{x}_{sd} R + (\dot{x}_{sd} - \dot{x}_s) \frac{Bs + K}{s} \Rightarrow \\ \Rightarrow F_{sc} &= \dot{x}_{sd} R + \frac{Bs + K}{s} \left[\dot{x}_{sd} - \frac{s}{m_s s^2 + Ds + Bs + K} \left(\dot{x}_{sd} \frac{Bs + K}{s} - F_d \right) \right] = \\ &= \frac{Bs + K}{s} \left(F_d \frac{s}{m_s s^2 + Ds + Bs + K} \right) \Rightarrow F_{sc} = F_d \frac{Bs + K}{m_s s^2 + Ds + Bs + K} \quad (7)\end{aligned}$$

$$\begin{aligned}
 F_{sc} &= \dot{x}_{sd}R + \frac{Bs+K}{s} \left[\dot{x}_{sd} - \frac{s}{m_s s^2 + Ds + Bs + K} \left(\dot{x}_{sd} \frac{Bs+K}{s} - F_d \right) \right] \Rightarrow \\
 &\Rightarrow F_{sc} = \dot{x}_{sd}R + \frac{Bs+K}{s} \left(\dot{x}_{sd} - \dot{x}_{sd} \frac{Bs+K}{m_s s^2 + Ds + Bs + K} \right) \Rightarrow \\
 &\Rightarrow F_{sc} = \dot{x}_{sd} \left[R + \frac{Bs+K}{s} - \frac{(Bs+K)^2}{m_s s^3 + Ds^2 + Bs^2 + Ks} \right] \quad (8)
 \end{aligned}$$

From (7) and (8) equations we get the following transfer function.

$$F_{sc} = F_d \frac{Bs+K}{m_s s^2 + Ds + Bs + K} + \dot{x}_{sd} \left(R + \frac{Bs+K}{s} - \frac{(Bs+K)^2}{m_s s^3 + Ds^2 + Bs^2 + Ks} \right) \quad (9)$$

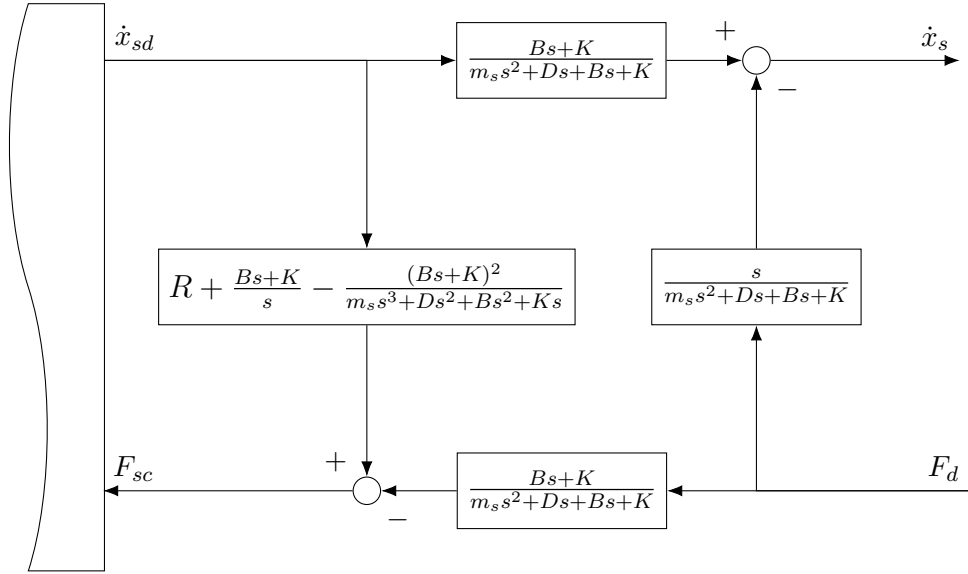


Figure 3.9: The final simplified block diagram of slave only impedance matching controller.

Using the same methodology we derive the master complete system, simplified into four transfer functions. The resulting block diagram is similar to slaves, with the exception of a few signs caused by differences in the "adding" parts of the diagram. We will shorten the procedure in order to not repeat the same operations. As before, we assume that $\dot{x}_m = \mathcal{L}\{\dot{x}_m\}$, $\dot{x}_{md} = \mathcal{L}\{\dot{x}_{md}\}$, $\dot{F}_m = \mathcal{L}\{\dot{F}_m\}$, $\dot{F}_{mc} = \mathcal{L}\{\dot{F}_{mc}\}$ and $\dot{F}_h = \mathcal{L}\{\dot{F}_h\}$ **only** symbolically.

$$\dot{x}_m = (F_h - F_m) \frac{1}{m_m s} = -\frac{F_m}{m_m s} = \frac{-\dot{x}_m D - F_{mcon}}{m_m s} \Rightarrow \frac{\dot{x}_m}{F_{mcon}} = -\frac{1}{m_m s + D} \quad (10)$$

$$\begin{aligned}
 \dot{x}_m &= (F_h - F_m) \frac{1}{m_m s} = \frac{-\dot{x}_m D - F_{mcon}}{m_m s} + \frac{F_h}{m_m s} = \frac{-\dot{x}_m D}{m_m s} + \frac{F_h}{m_m s} \Rightarrow \\
 &\Rightarrow \frac{\dot{x}_m}{F_h} = \frac{1}{m_m s + D} \quad (11)
 \end{aligned}$$

From (10) and (11):

$$\dot{x}_m = -\frac{F_{mcon}}{m_m s + D} + \frac{F_h}{m_m s + D} = \frac{1}{m_m s + D} (F_{mcon} + F_h) \quad (12)$$

While keeping our focus at the transfer functions of $\dot{x}_m$, we simplify further the system.

$$\begin{aligned}\dot{x}_m &= -\frac{F_{mcon}}{m_ms + D} + \frac{F_h}{m_ms + D} = -\frac{1}{m_ms + D} \frac{Bs + K}{s} (-\dot{x}_{md} + \dot{x}_m) = \\ &= -\frac{Bs + K}{m_ms^2 + Ds} (-\dot{x}_{md} + \dot{x}_m) \Rightarrow \dot{x}_m = \dot{x}_{md} \frac{Bs + K}{m_ms^2 + Ds + Bs + K}\end{aligned}\quad (13)$$

$$\begin{aligned}\dot{x}_m &= -\frac{F_{mcon}}{m_ms + D} + \frac{F_h}{m_ms + D} = \frac{1}{m_ms + D} \left[-\frac{Bs + K}{s} (\dot{x}_{md} - \dot{x}_m) + F_h \right] = \\ &= \frac{1}{m_ms + D} \left[-\dot{x}_m \frac{Bs + K}{s} + F_h \right] \Rightarrow \dot{x}_m = F_h \frac{s}{m_ms^2 + Bs + Ds + K}\end{aligned}\quad (14)$$

Using (13) and (14) we have the following.

$$\dot{x}_m = \dot{x}_{md} \frac{Bs + K}{m_ms^2 + Ds + Bs + K} + F_h \frac{s}{m_ms^2 + Bs + Ds + K}\quad (15)$$

At last, we end our simplification procedure by analysing F_{mc} .

$$\begin{aligned}F_{mc} &= -\dot{x}_{md}R + (\dot{x}_m - \dot{x}_{dm}) \frac{Bs + K}{s} \Rightarrow \\ \Rightarrow F_{mc} &= -\dot{x}_{md}R + \frac{Bs + K}{s} \left[\dot{x}_{md} \frac{Bs + K}{m_ms^2 + Ds + Bs + K} + F_h \frac{s}{m_ms^2 + Ds + Bs + K} - \dot{x}_{md} \right] \Rightarrow \\ &\Rightarrow F_{mc} = F_h \frac{Bs + K}{m_ms^2 + Ds + Bs + K}\end{aligned}\quad (16)$$

$$\begin{aligned}F_{mc} &= -\dot{x}_{md}R + \frac{Bs + K}{s} \left[\dot{x}_{md} \frac{Bs + K}{m_ms^2 + Ds + Bs + K} + F_h \frac{s}{m_ms^2 + Ds + Bs + K} - \dot{x}_{md} \right] \Rightarrow \\ &\Rightarrow F_{mc} = \dot{x}_{md} \left[-R - \frac{Bs + K}{s} + \frac{(Bs + K)^2}{m_ms^3 + Ds^2 + Bs^2 + Ks} \right]\end{aligned}\quad (17)$$

From (16) and (17) equations we get the transfer function.

$$F_{mc} = F_h \frac{Bs + K}{m_ms^2 + Ds + Bs + K} + \dot{x}_{md} \left[-R - \frac{Bs + K}{s} + \frac{(Bs + K)^2}{m_ms^3 + Ds^2 + Bs^2 + Ks} \right]\quad (18)$$

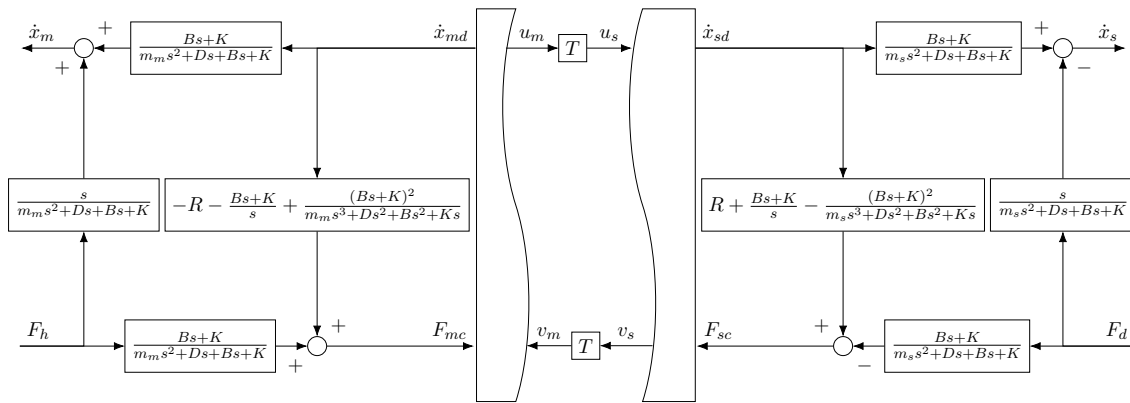


Figure 3.10: Final complete simplified block diagram of the impedance matching method.

Chapter 4

Experiments

In this chapter, we define the final complete teleoperation architecture. We test its capabilities, as well as the impact every subsystem has onto the complete scheme, through different means. We also go in depth about the systems used for these tests, the values that were chosen for the variables of the system and the reasons behind each of these selections. Finally, we will showcase and discuss a plethora of graphs, showcasing the systems behavior under different circumstances.

4.1 The system

The complete system we created is divided into two parts, the software and the hardware part. The software part of the system is used to implement the complete teleoperation scheme, the environment the robot is operating in, as well as the communication interface of the haptic device the human operator is using to interact with the rest of the system. The hardware part is a three degree of freedom haptic device, the operator can use to apply and receive forces to and from the environment.

Starting with the haptic device, we use the Novint Falcon, a 3 DOF haptic controller connectable with a standard USB. It has the capability to measure the position and velocity of the "clutch" (see Figure 4.1) in the three dimensions, in addition to the force applied from and to the operator. The disadvantage of this haptic device is the relative small workspace, only 20cm in the biggest section of its conical shape, and the extra two degrees of freedom that we can not use with the current control scheme and will need to be locked.

As for the interface between the haptic controller and the rest of the system, a commonly used in the robotics sector operating system, ROS 1, is used. ROS 1 stands for "Robot Operating System" and it uses a combination of topics and nodes to represent information channels and robotic systems respectively. We used the first iteration of ROS, which is only supported on Linux based operating systems, due to its immense catalog of supported libraries, one of which "libnifalcon" is used to link the two systems. Unfortunately this library was tested before only up to an old 2018 ROS 1 version, Melodic Morenia, released for the Ubuntu 18.04 operating system. Testing it anew in the current version of ROS 1, Noetic Ninjemys, released in 2020 for Ubuntu 20.04, seemed to work without any problem and will be used for the implementation of the system. The end goal of this operating system is to transfer the force and velocity signals between the haptic device and the rest of the teleoperation system.

The complete teleoperation system is comprised of the three distinct sub-systems we analysed in the previous parts of this thesis, a position tracking system with the "Adjusting the wave command" approach, an on-line tuning system and the impedance matching system to deal with any wave reflections that appear during the control sequence. We



Figure 4.1: The Novint Falcon haptic controller.

might think that we need to prove the passivity argument along with the characteristics of each subsystem, as by combining all three we do not know the new behavior of the system. In their work (Günter Niemeyer and Slotine 2004) argue that the position tracking system and the impedance matching method does not affect the passivity criteria and can be used in any wave-based teleoperation system, something we will examine experimentally in the following sections of the thesis, along with their combined effectiveness. On-line tuning is not an autonomous system and does not change our architecture, it is an inherent function of the wave variables and does not depend on other changes in the model, so it preserves the passivity criteria of the rest of the system. However, we will test its characteristics and its relationship with the rest of the subsystems.

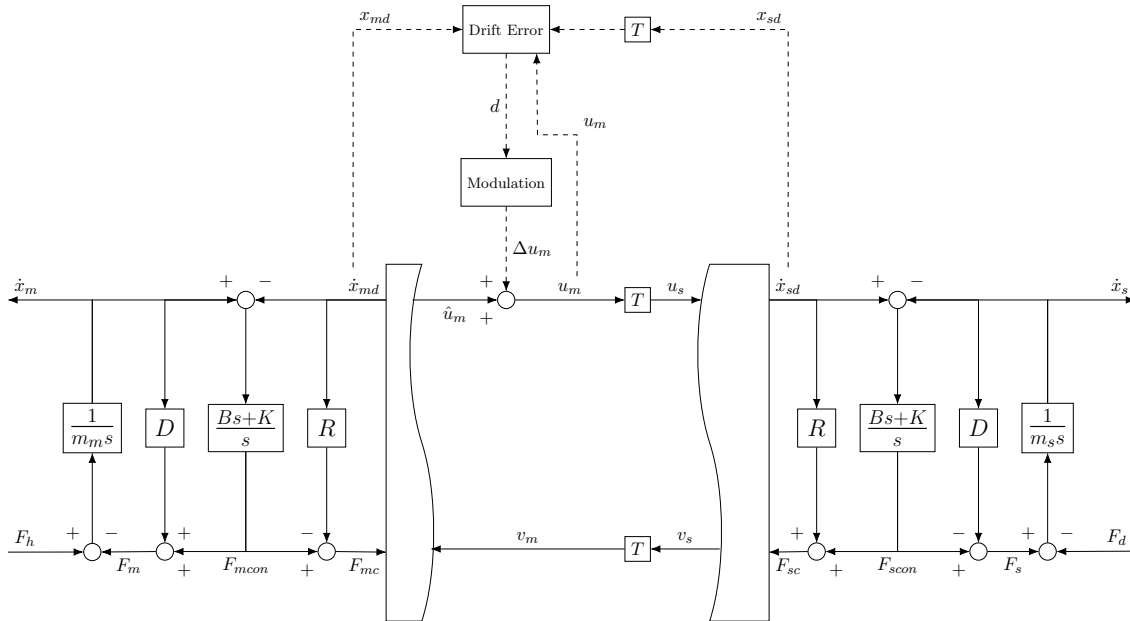


Figure 4.2: The complete teleoperation control scheme.

The behavior of the scheme in use, is pretty intuitive. Two of the three axes of the haptic device are locked with a basic PID controller because the architecture is capable of

single degree of freedom control. The only "free" axle is the forwards and backwards one, or the y axle. In the beginning of the procedure a homing operation needs to take place, in order to calibrate correctly the three encoders of the device, after which the "clutch" is automatically locked in the home position in the middle of the workspace. In order to start the procedure the middle button of the clutch needs to be pressed, after which movement in the y axle is unlocked and the full teleoperation is active. When the user deems to end the control sequence, the middle-upper button can be pressed to close the system and display the trajectory and force information of the complete operation in graphs.

The delay chosen in all of our following experiments is 1sec round trip time (RTT). This selection is inspired by the average delays that govern the communications between the International Space Station (ISS) and the DLR-GSOC ground station in their space telerobotics missions. These missions conducted in the years between 2005 and 2015, and used force feedback as a main feature of their system, very similar to ours. So 1sec round trip delay seems a reasonable choice with some head room for greater distances or worse communication mediums, but we tested the system from zero, up to two seconds round trip delay, with great success.

Experiment	Type	RT-Delay	P.Loss	Bandwidth
ROKVISS	ISS Link	20-30 ms	0.1%	256Kbps/4Mbps
ARTEMIS	GEO-Sat	620 ms	5.8%	4 Mbps
ASTRA	GEO-Sat	540 ms	2.6%	4 Mbps
KONTUR-2	ISS Link	20-30 ms	0.1%	256Kbps/4Mbps
K2 Training	Internet udp	65 ms	10%	10Mbps

Figure 4.3: Main DLR space telerobotics missions communication parameters. P.Loss:Packet Loss; RT:Round Trip (Artigas and Hirzinger 2016).

The experiments that follows can be slit up into three parts. We decided to test the system firstly in a "sanitary" environment which is a simulation, then step it up by introducing a real life haptic devise for the operator to control, and finally alter the behavior of the master sub-system in order to better represent real life tasks. The slave and master model used in every teleoperation control scheme we analysed so far, was that of a simple mass. We try to improve it by adding friction and bring the resulting movements closer to reality. We will do three main tests for each part, firstly with only the basic teleoperation system constructed from the impedance matching methodology (see Figure 3.6), then we will add one by one the online tuning and position tracking capabilities and finally combine all the systems together. Our position tracking approach of choice is the "Adjusting the Wave command" method. After each test we will briefly discuss the resulting movement, the observable drawbacks of the system tested if any and the specific values chosen for the variables of the system.

4.2 Modeling and Simulation

A simulation is a great way to experiment on a system because its a risk free environment, cheaper, and some times a lot faster. In this thesis, we simulated the complete teleoperation system along with the communication channels, the environment the slave robot is operating in and the controlling behavior of the operator. In the beginning we run our experiments with master and slave represented as simple masses but we then improve them by adding friction. The simulated environments consists of free space and free space with an immovable object at some distance. The purpose of the free space environment is

to examine the movement of slave under a simple "move" control sequence, while the existence of a wall in the second environment serves the purpose of testing the force feedback capabilities of the system, as well as the force the slave robot applies into the environment.

The 1sec round trip delay applied to the system displays no jitter, so the opposite side will receive the messages sequentially after the initial delay passes, before which, in the case of slave, it remains dormant.

4.2.1 Simulation in free-space

The Basic Model

We begin our experimental procedure with the basic teleoperation model depicted in Figure 3.6. Because we are in a simulated environment we can create any input we want for the system. For the first experiment we chose a sine wave as a force input of the human operator.

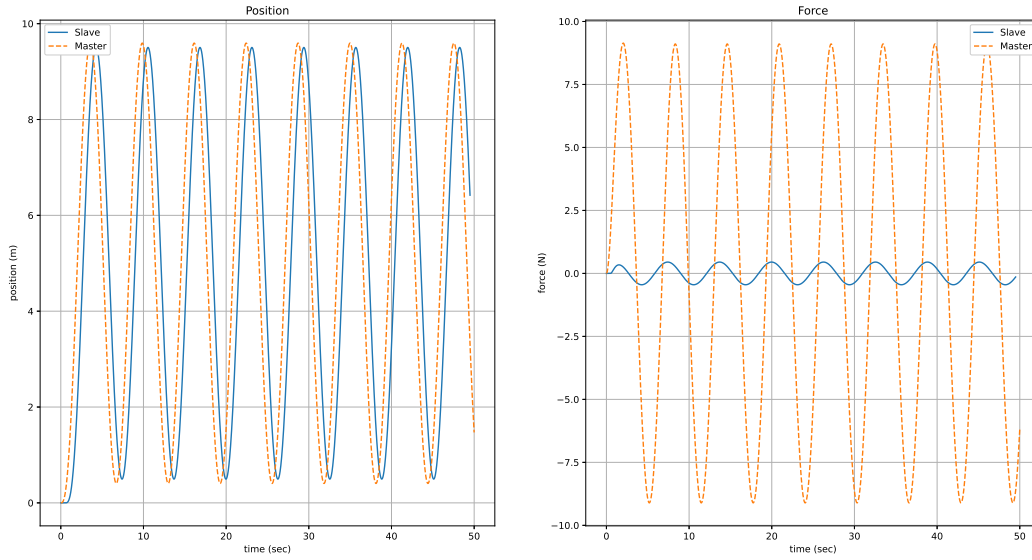


Figure 4.4: Experiment with the simulated basic control scheme in free space .

In the resulting movement we observe that the slave movement trails that of the master as it should, the force applied by the slave does not match that of the master because we are in free motion and we have not hit an object. The small sine behavior of the slave force is caused by the "desire" of the slave system to overcome its own inertia and match its velocity to the velocity of the master. In very simple conditions like this, where the task consists of only free movement and has no friction or huge delays, the basic wave reflection countering system is sufficient and adding the position matching and specifically online tuning subsystems is not necessary. But we want to test their effects on the behavior of the system in order to better understand them.

Online Tuning

We continue on with the addition of the online tuning subsystem. As we mention before, this system is not necessary or even useful in simple freespace movements. Its is capable of stiffen or relax the systems behavior by increasing or decreasing the wave impedance, required for contact operations. We can not relax the system any further as a correct tune for the freespace movement requires the wave impedance as low as possible, so we can demonstrate the effects of increasing the wave impedance unnecessarily in the middle of an operation.

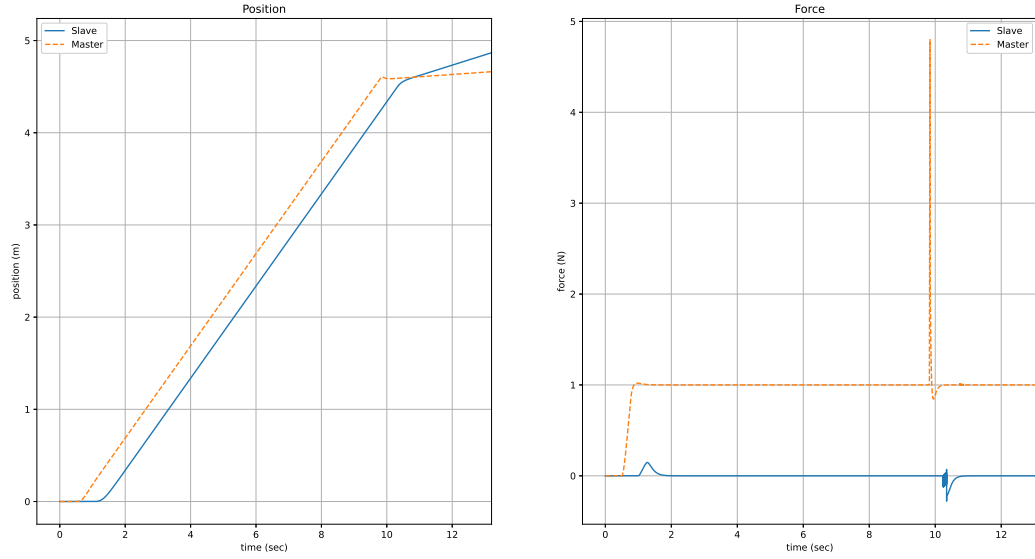


Figure 4.5: Experiment with the simulated online tuning control scheme in free space.

As we can see, by increasing the wave impedance we stiffen up the system and almost stop its movement. Despite that, we do not see an increase in the slave or the master force. The master force remains the same as the human operator exerts exactly the same force as before so only the master velocity is decreased. The slave force also remains the same at zero because we are in a "clean" simulated environment and there are no extra forces like friction.

Position Tracking

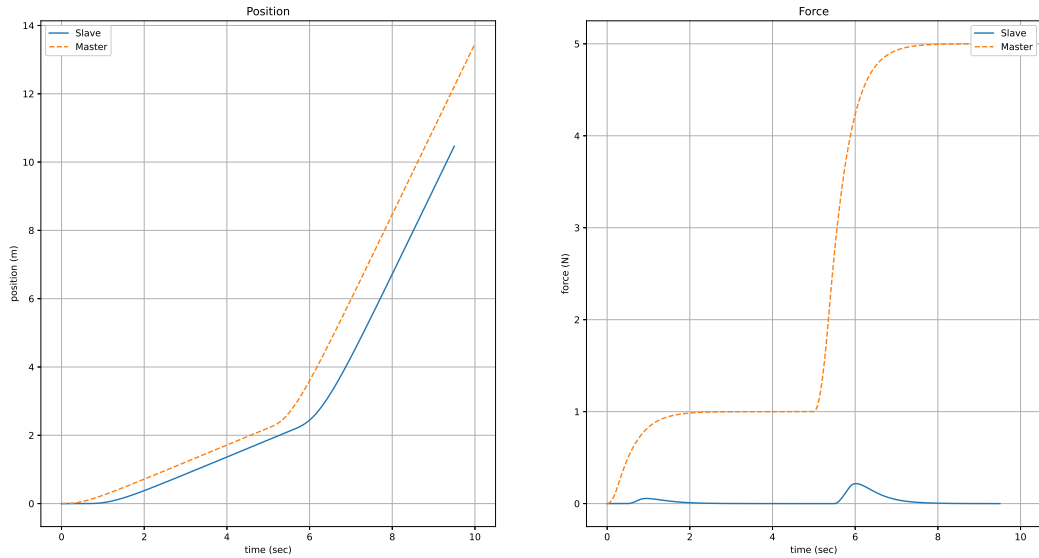


Figure 4.6: Experiment with the simulated "adjusting the wave command" control scheme in free space.

Position tracking is another subsystem that in a simulated, without friction or noise environment is not needed. The generated movement has the correct behavior and does not display any problems, but this is due to the impedance matching controller and not the wave correction subsystem.

Combined Model

Finally, we combine the three components, impedance matching, online tuning and position tracking, to get the final system with the following behavior.

The final movement carries the problem of an unnecessary stiffness caused by the online tuning system. There is also a difference with the previous online tuning case, where we did not include position matching in our scheme, which is that after the point where we stiffen up the system the master and slave velocities match each other at almost zero. This is a result of the position tracking component trying to match the master and slave positions and not let them increase with different rates like the previous experiment. In conclusion, in an environment like this one with a simple movement task, the online tuning subsystem is not useful and the position tracking component does not have any position drift to correct, so is also not necessary.

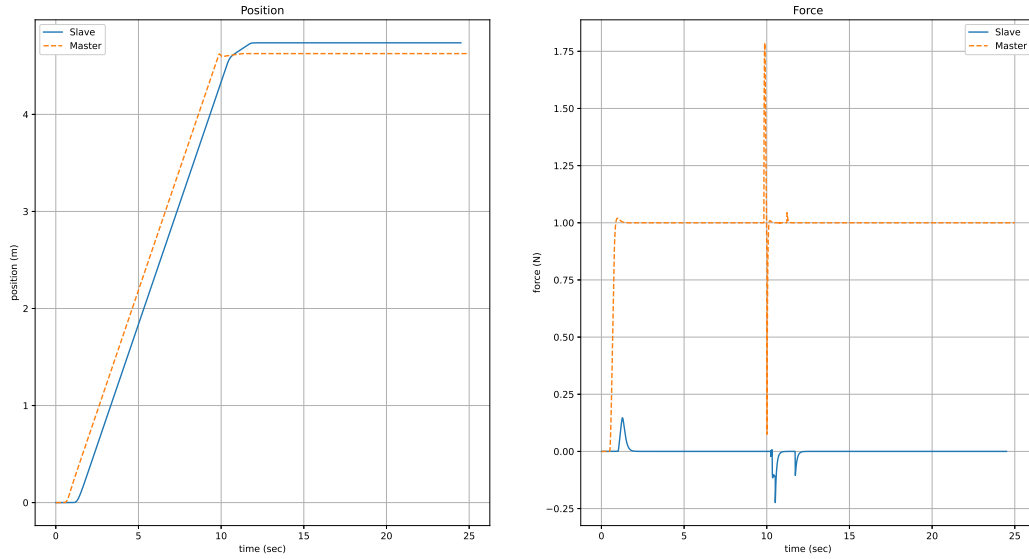


Figure 4.7: Experiment with all three simulated subsystems combined (impedance matching, position tracking, online tuning) in free space.

4.2.2 Simulation in free-space with an obstacle

Basic Model in free-space with a Wall

Another very common environment a robot can encounter, besides a freespace, is an environment with an immovable object in it. In this case the task of the robot might be a "push" type where the robot needs to make contact with the object and transfer the force the operator is applying to the haptic device, directly to the object.

To test this case we start with the basic model and use as force input from the operator a two step function.

As we can see, in the first step the robot operates in freespace so the slave movement is simple and tracks that of the master as expected. At the second step where the slave robot encounters the obstacle we observe some oscillations in the master position and both the master and slave forces, but after a small amount of time they both stabilize.

The oscillations in force, as well as the masters position, happen due to the instant change in environment that is a "hit" with an object and the passive nature of the system. The small increase that happens in the slave force in the beginning of every step is a result of the controller trying to accelerate the slave to match the velocity of the master.

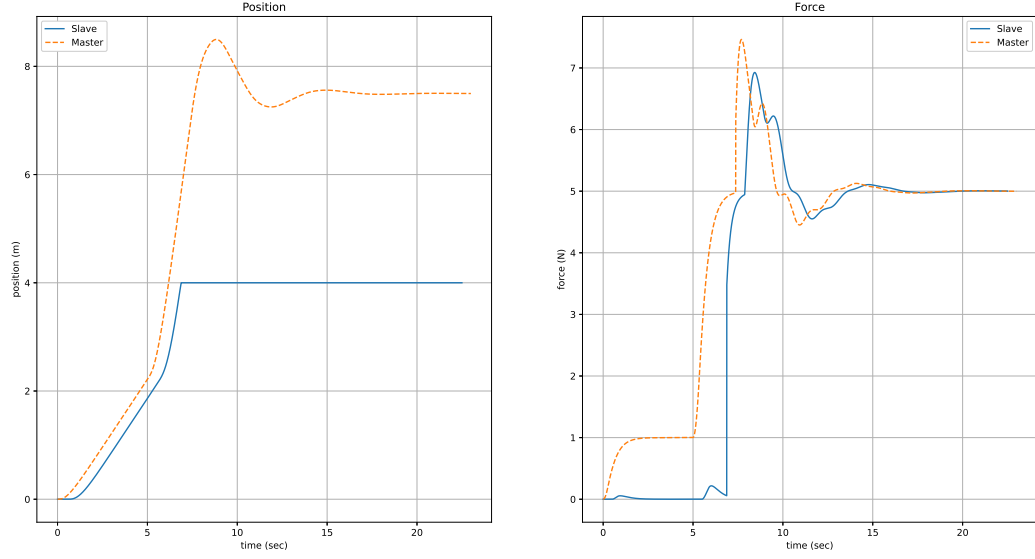


Figure 4.8: Experiment with the simulated basic control scheme in free space with an immovable object.

Online Tuning in free-space with a Wall

In the next step of our experiments we introduce the online tuning subsystem. The online tuning component is very important in order to correctly carry through a "hit and push" task, as we need to make the controller stiff and transmit forces, not velocities. As we analysed in the "Position Drift" chapter, if we increase the wave impedance thought the online tuning process, just before the slaves contact, we negate the initial spike in position and reduce the difference in the two position, caused by the delayed reaction of the master robot.

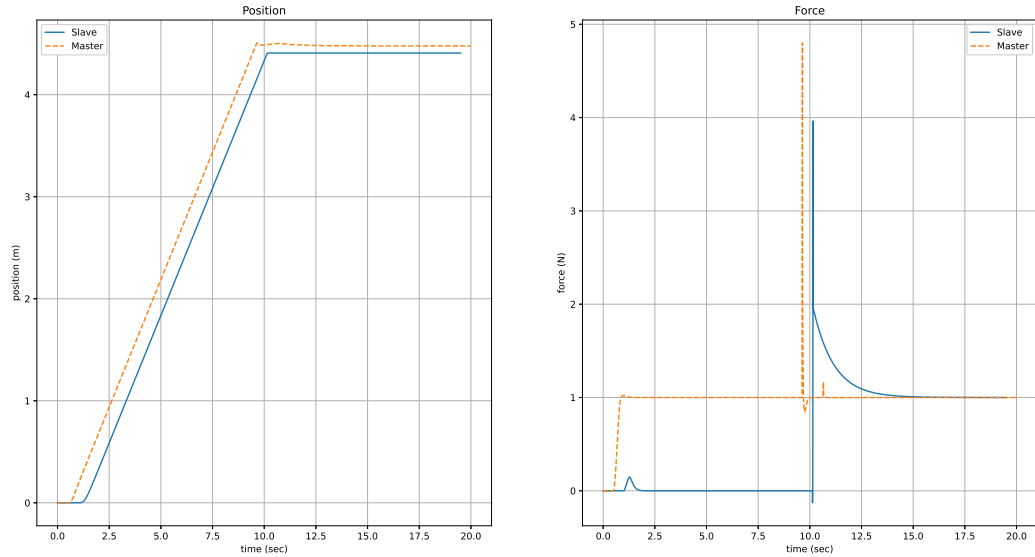


Figure 4.9: Experiment with the simulated online tuning control scheme in free space with an immovable object.

We can see this behavior in the above Figure as the master travels a very small distance beyond the slave hit position. The slave force jumps up to match the master force and apply it to the obstacle, so this system carries all the necessary attributes to carry through

this type of tasks.

Position Tracking in free-space with a Wall

By removing the successful online tuning subsystem we can test the impact a position tracking component has on the behavior of the system in this environment.

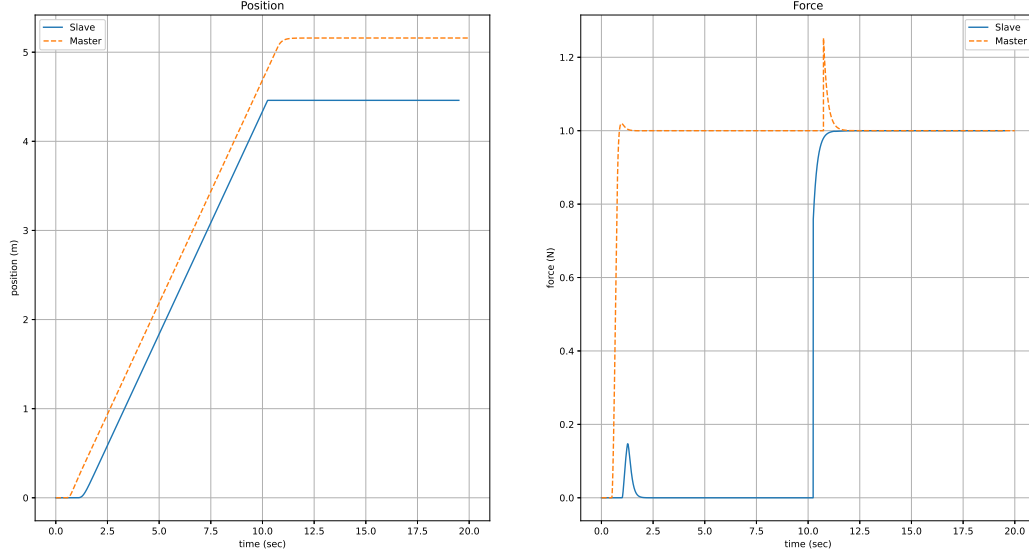


Figure 4.10: Experiment with the simulated "adjusting the wave command" control scheme in free space with an immovable object.

We need to compare the resulting movement to that of the basic system in order to understand the necessity of this component. Observing the Figure, we see that the master has moved somewhat far from the slave hit position while carrying the same speed, but unlike the basic model movement, the master velocity returns to zero after the signal carrying the hit information reaches the master robot. We see no oscillation neither in the master position nor in the slave and master forces. After the impact, the slave force increases to match the masters smoothly and quickly, while the master force has a small spike in order to stop its movement.

So in cases where the online tuning method fails or is not implemented the position tracking subsystem is capable to transfer the event of contact from the slave to the master and apply the correct force to the environment after the hit.

Combined Model in free-space with a Wall

Combining all the subsystems together we get the following results. We observe a movement that carries all the characteristics of the component included and the correct conduct of the task. Some oscillations are present after the contact event but they are caused by the imperfect online tuning impedance values and their rate of increase, as well as the natural oscillations in the wave variables.

As a consequence of the higher velocities, present in the second step of the motion, the difference in the master and slave position after the stabilization is perceived large, but it is in the same lines as the previous experiments as we tested the system with the same input as the position tracking and online tuning experiments. Noteworthy is the fact that, even though we do not display the results here, the resulting movement was exactly the same as in the Figure 4.9, which means that the position tracking subsystem did not contribute in any major way and is probably redundant in this exact scenario.

To conclude this segment we can say that the complete teleoperation controller is capable of handling tasks that involve movement in freespace as well as "push-pull" missions with an object existing in the same freespace, under the effects of increased delay.

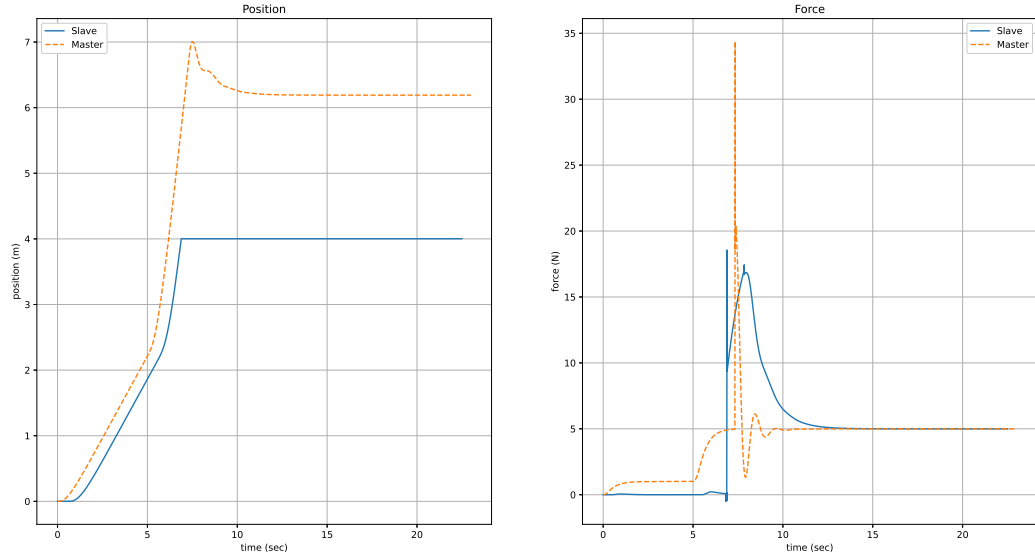


Figure 4.11: Experiment with all three simulated subsystems combined (impedance matching, position tracking, online tuning) in free space with an immovable object.

4.2.3 Improved master and slave models

After we tested the complete system in a simulation and made sure all the subsystems work as intended, we decided that a simple mass model of the master and slave robotic system is not realistic enough and does not represent a real life scenario. As a result we added friction to the slave model, which consists of a viscous friction depended on the velocity of the robot, and a linear model of the Coulomb friction. Even with friction added, the system is vastly simplified compared to a real life model, but in contrast to a simple mass model, adding a viscous and coulomb friction improves it significantly.

We rerun all the previous experiments to test the new models and see the difference adding friction makes to the movement of the robots. In order to prevent repetition, we will present only a few example movements we experimented on and discuss the overarching themes and behaviors that are present.

We begin as always with the simulated freespace environment and simple movements. Using the basic teleoperation scheme the effects of friction become visible, the master and slave velocities do not match each other with the slave velocity being slightly less in free movement. This is the effects of friction as it "resists" the slave system in matching its velocity to the desired velocity of master (see Figure 4.12). We also observe a larger initial spike in the master force at the beginning of each movement, which can be explained by the effects of the Coulomb friction in closer to zero velocities.

The online tuning subsystem, as we analysed extensively in the past segment, is not usable in this kind of movements, so the resulting behavior after increasing the wave impedance through the online tuning system is not desirable. None of the friction characteristics are eliminated with this method.

However, the position tracking component manages to diminish a major effect of friction in the resulting movement. The divergence between the masters and slaves velocities in a steady state movement is eliminated after a few seconds of stabilization. As we can see in Figure 4.13, at the start of each step the two velocities are not identical, but after

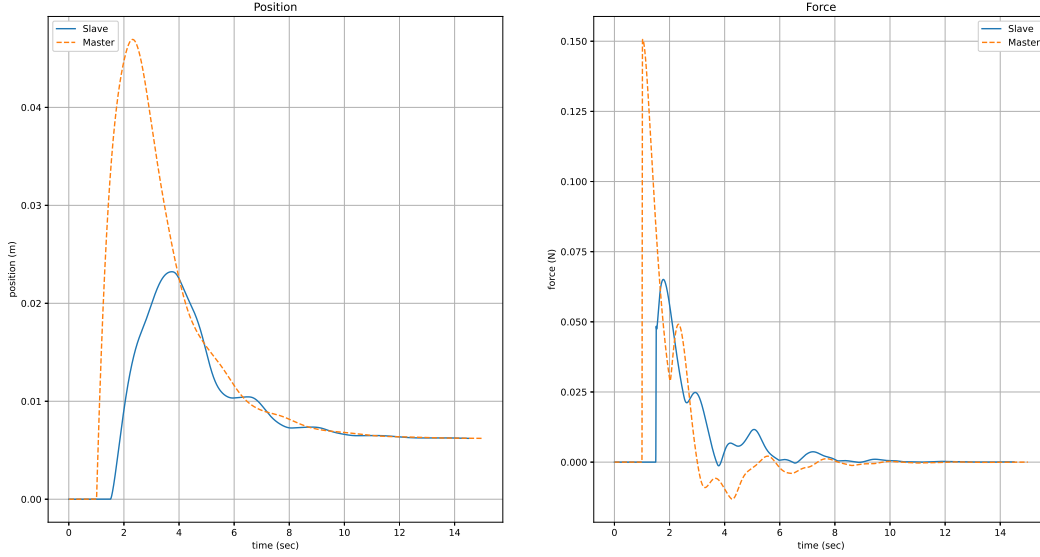


Figure 4.12: Experiment with the simulated basic control scheme in free space with an immovable object and friction.

some seconds, the master velocity decrease to match that of the slave achieving equality. An also important detail is the fact that even though we have free movement and the slave is not in contact with any obstacle in the environment, the slave force is not zero even after the initial inertia it needs to overcome. This phenomenon is caused by the friction and to understand it we can visualize friction as a small in mass object that the slave robot needs to "push" every time it moves. So if slave is pushing a small object it is considered a contact task and it needs to transmit a percentage of the master applied force into the object, which is exactly the behavior we observe.

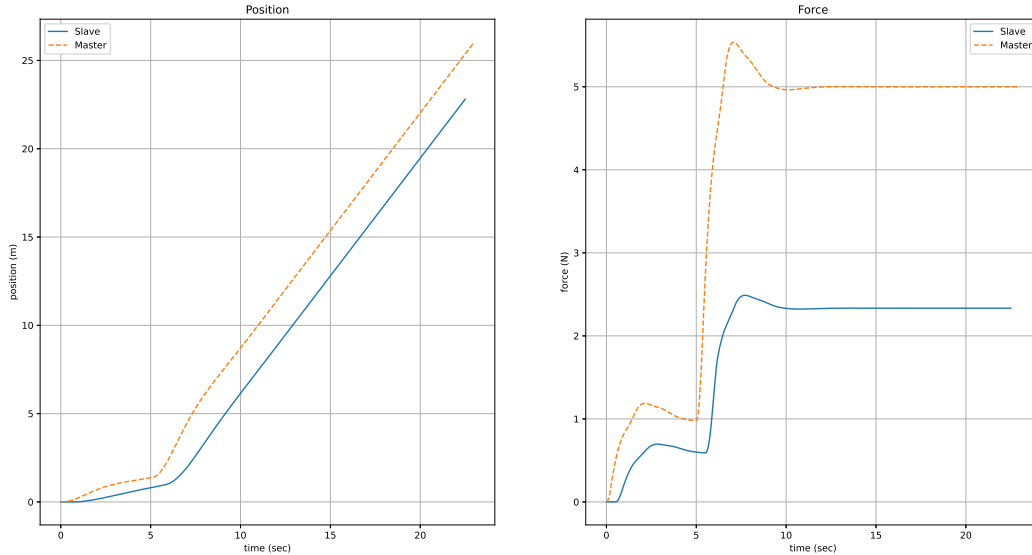


Figure 4.13: Experiment with the simulated "adjusting the wave command" control scheme in free space with an immovable object and friction.

The complete controller combines the behaviors of the 3 subsystems. It displays the correct behavior, without some friction characteristics due to the position tracking component, up until the online tuning point where as we discussed before the system becomes

unnecessarily stiff resulting into smaller movements.

Moving on to a completely immovable obstacle existing in the freespace to test the extreme scenario of a "hit". We want to see a fast stabilization in the masters position after the slave contact with the environment and a exact transmission of force from the master systems to the obstacle.

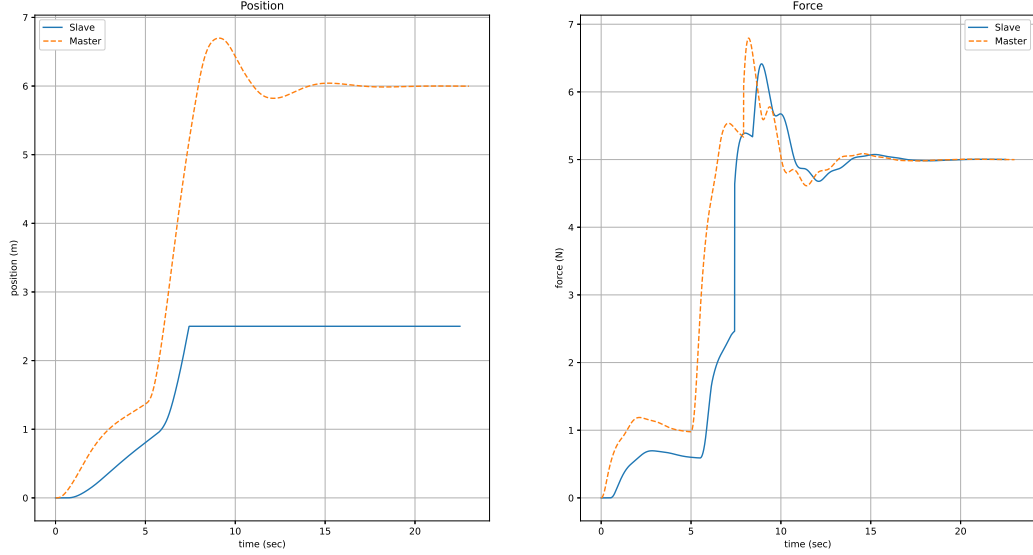


Figure 4.14: Experiment with the simulated "adjusting the wave command" control scheme in free space with an immovable object and friction.

In the freespace portion of the movement there are no changes from the previous experiments as the only change is the existence of the wall in the environment. The basic model is capable of handling the contact event successfully. The oscillations in the both the master position and forces dissipate quickly and the slave force increases to match exactly that of the master after the contact. The only existing minor problems is the deviation in position and velocity between master and slave, that is caused by delay and friction respectfully.

This time the online tuning functionality of the system is useful as we can stiffen the system right before the contact event and decrease the position difference between master and slave that happen due to the delayed contact information.

The position matching does not impact the behavior of the system after the impact with the obstacle, but it has the same benefits we observed earlier in negating the effects of friction in the steady state movement part of the experiment.

With all the subsystems together we have a competent control system that can handle both freespace and contact operations under large delays. The improved model with added friction while it changes the behavior of the robot, the controller is capable of almost negating its effects through the position tracking component in the steady state movement, while in contact, friction has almost no impact. A simulation "sterilizes" the experiments and is not the best depiction of a real life scenario, but it displays and helps us understand the basic behavior of the combined system, as well as each subsystem.

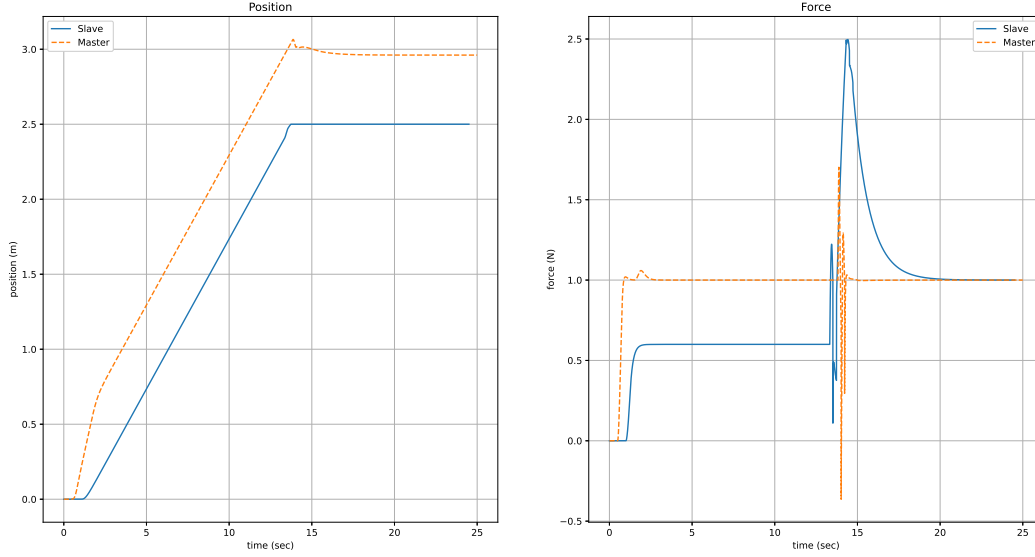


Figure 4.15: Experiment with all three simulated subsystems combined (impedance matching, position tracking, online tuning) in free space with an immovable object and friction.

4.3 Experimental Implementation

4.3.1 Novint Falcon

The next step in our set of experiments is to upgrade from a simulation to a more complete and realistic model. For this reason we introduce the Novint Falcon haptic controller depicted in Figure 4.1 to replace the simple mass model of the master subsystem. Because adding a physical device in the place of a much simpler simulated one is a big change, we will examine the new case with a simple slave model and we will add friction afterwards. In the following graphs we will observe some noise which is caused by the unstable nature of the human hand and some stiffness to small changes in the response of the master force, also caused by the inability of a human hand to detect small changes in pressure. Lastly, due to the existence of the extra two PID controllers on the haptic device, trying to prevent any movement in the x and z axle, we will detect very small in duration spikes in stiff movements like touching the immovable wall.

Basic Model in free-space

Starting with the basic teleoperation scheme in a free space environment we test the systems ability to simply match the velocity of master to the velocity of slave in the presence of delay.

In this first experiment we observe an initial shift in the masters position. This is due to the pressure required by the operator to press the start button on the Falcons grip and we will see it in every falcon experiment. The wave impedance was tuned somewhat high so the forces involved are greater both in slave and master sub-systems. We also observe a small force applied by the slave robot every time it moves which we can blame it on the slaves effort to overcome inertia. Master applies force through the haptic device to the operators hand to simulate the reflected forces involved in movement of a mass and we notice that when there is no movement, both slave and force forces revert to zero. The slave movement is trailing that of masters and the whole movement is correct.

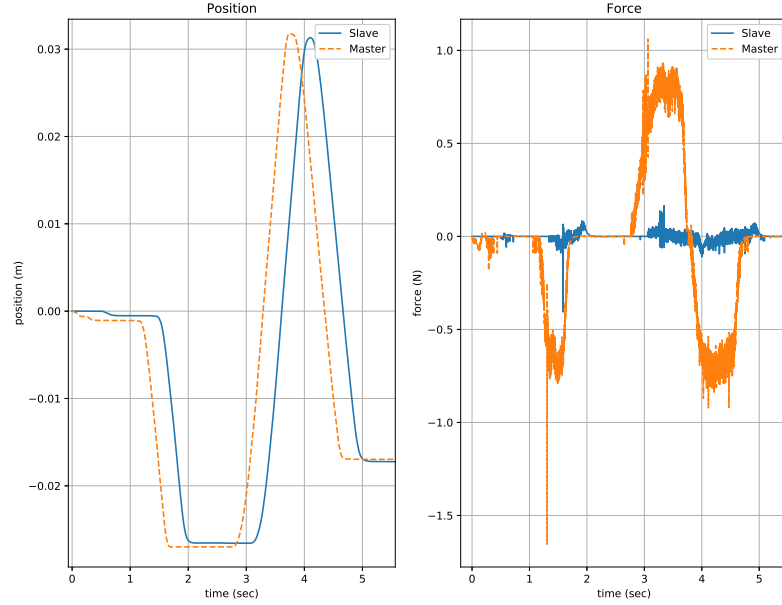


Figure 4.16: Experiment with the basic control scheme and Novint Falcon in free space.

Online Tuning in free-space

Moving on, we add the online tuning behavior into our system, but because we are operating in free space, tuning up the wave impedance is not required. However we can think of an operator observing that the required task comprises only of free movement and tune the wave impedance down to enhance the systems agility. So in our next experiment we tune down the wave impedance as well as the systems bandwidth λ in order to satisfy the impedance matching limitations (reminder: $b \geq \lambda m$).

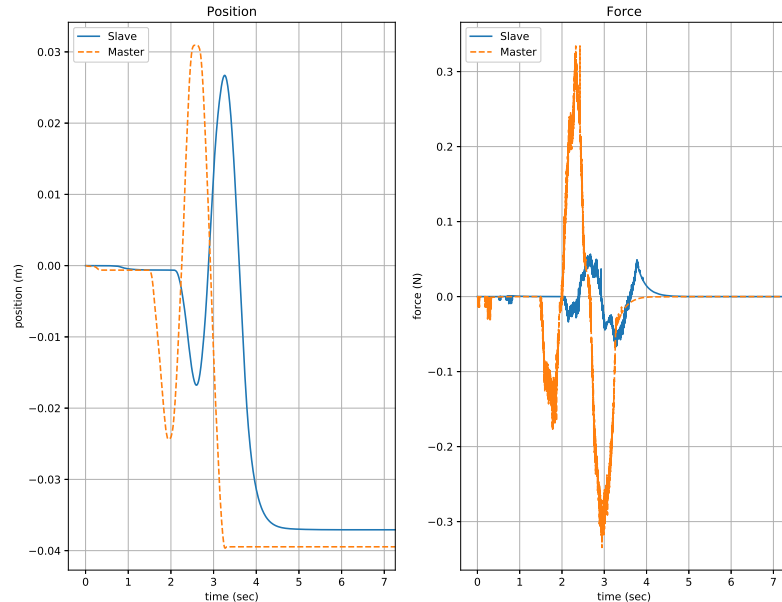


Figure 4.17: Experiment with the online tuning control scheme and Novint Falcon in free space.

On this new experiment we see smaller forces taking place due to the lower value of the wave impedance. The noise in the forces included has been reduced due to the fact that the system is less stiff with lower tuned wave impedance values. We also find a small damping

effect in the movements of the slave robot which was caused by the necessary lowering of the systems bandwidth. Despite that, the system works in its current environment, with the slave movement tailing the masters.

Position Tracking in free-space

The next subsystem to add in the basic teleoperation scheme is that of the "Adjusting the wave command" method. To test its effectiveness we remove the online tuning approach, which means higher wave impedance and bandwidth values, and run the test in a free space environment.

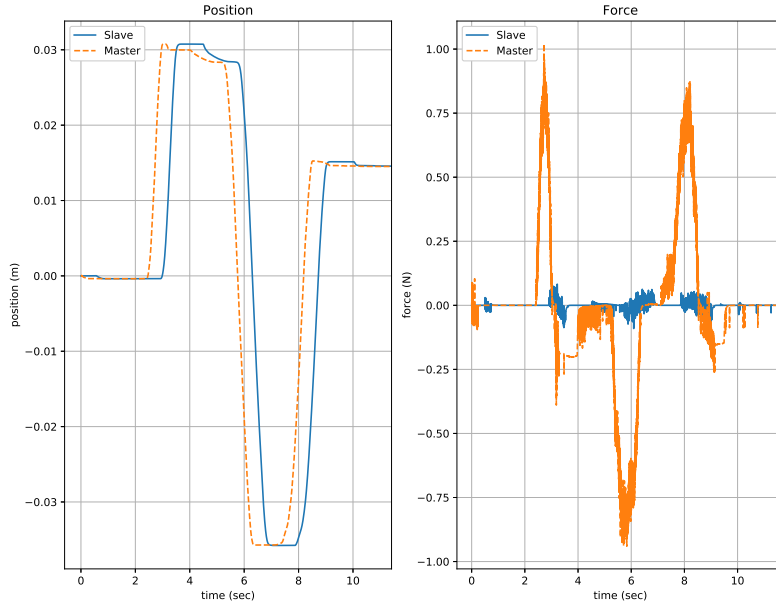


Figure 4.18: Experiment with the "adjusting the wave command" control scheme and Novint Falcon in free space.

Here we observe the return of higher force values and small, fast oscillations in their signal. Compared to the previous control schemes, we notice that the master and slave positions are identical at the end of each movement, which is a sign of the position tracking subsystem working as intended.

Combined Model in free-space

Lastly, we combine the position tracking and the online tuning system onto the basic impedance matching model, to create a system that deals with all the major problems of a wave-based teleoperation control experiment. We are expecting to see no wave reflections, or if exist, to dissipate quickly, an agile system, tuned for movement in free space, and we expect the slave and master positions to match after the end of each small movement.

In the resulting graph we discern all the characteristics of the aforementioned systems. We see matched slave and master positions after each movement, small forces with little to no noise and no wave reflections. The slave robot movements trace the movements of the haptic device, even though there is a damping effect due to the low tuned bandwidth of the system.

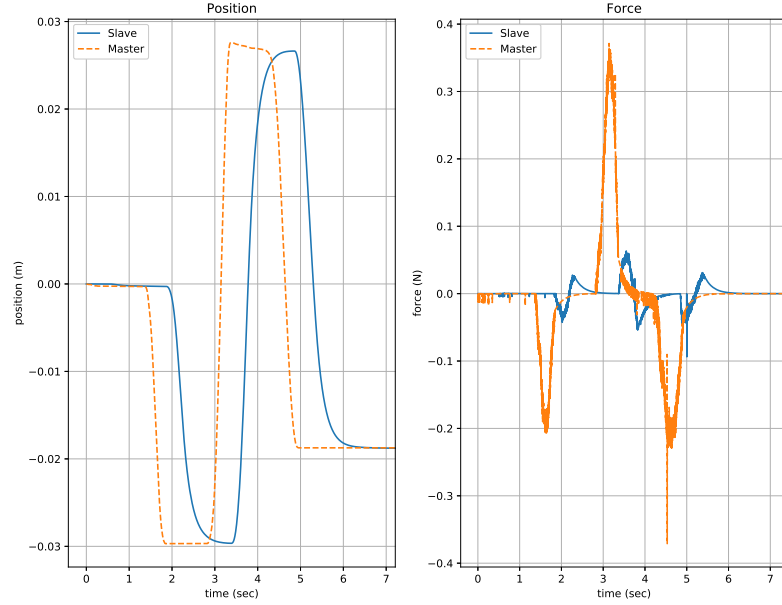


Figure 4.19: Experiment with all three subsystems combined (impedance matching, position tracking, online tuning) and Novint Falcon in free space.

Basic Model in free-space with a Wall

Also very important is the case of an immovable object or a wall existing in our freespace environment. When the slave robot touches the wall we need to transfer the force the operator is applying into the haptic device, into the wall, as well as return to the operator the forces the wall is applying to the slave robot. We will begin with the basic impedance matching model and will slowly add new capabilities to the system.

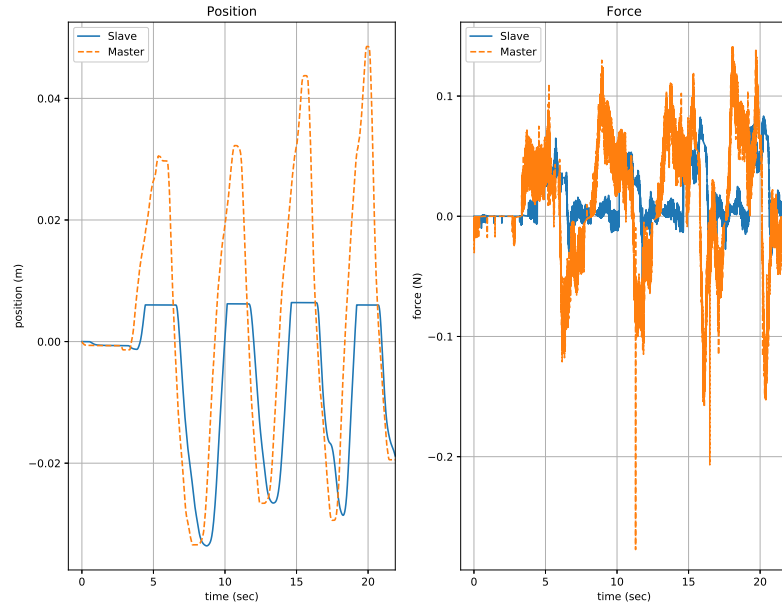


Figure 4.20: Experiment with the basic control scheme and Novint Falcon in free space with an immovable object.

The simulated wall was placed 16cm in front of the starting position. As we can see the slave robot stops when it "hits" the wall, but master only has a small decrease in

its velocity 0.5 seconds after the impact of slave. This is to be expected as the wave impedance is yet too low and the returning forces are not large enough to stop the human hand, effectively the operator keeps increasing the force applied to the controller and, despite the efforts of the teleoperator, prevents master from stopping.

Another noteworthy detail of the graph is that when slave makes contact with the environment, the force applied by master transfer perfectly to the obstacle the slave robot is facing, and when is free space movement only the inertial forces are observed, a main feature of the wave-based force feedback teleoperator we have been analysing.

Online Tuning in free-space with a Wall

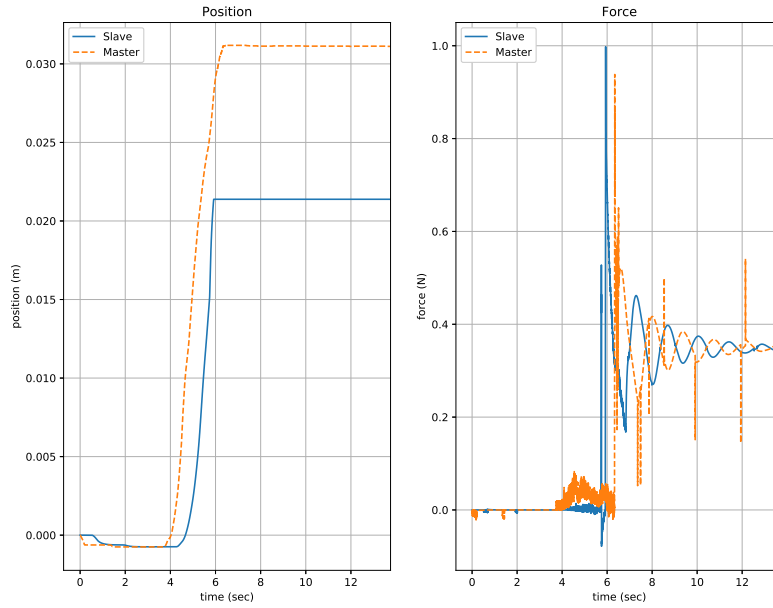


Figure 4.21: Experiment with the online tuning control scheme and Novint Falcon in free space with an immovable object.

The subsystem we need to correctly reflect the forces applied by the environment with a human operator is online tuning. We need to stiffen up the system prior to the impact in order to prevent big spikes of force reflection to appear, and retain this stiffness as long the slave robot pushes at the immovable object. As we analysed earlier, we can stiffen up the system by increasing the wave impedance. We chose to increase the wave impedance gradually 0.1cm before the impact with the wall and reduce it instantly to previous values when the slave robot stops touching the wall.

In the resulting graph we can observe that after the impact, the slave force rises to match that of the masters, and even though there are oscillations, they dissipate after a few seconds, so the system is stable.

At our tests we have noticed a problem with the current system, consisting of only the impedance matching and the online tuning subsystems. That problem is position decoupling and we observed it in the cases of repeatedly hitting the wall and then retreating (see Figure 4.22). In order to resolve this issue we will add position tracking capabilities to the system in the next step of our experiments.

Position Tracking in free-space with a Wall

Position tracking is the final subsystem that we need to add on top of the basic impedance matching controller. Similarly to the free space experiment, the system deals

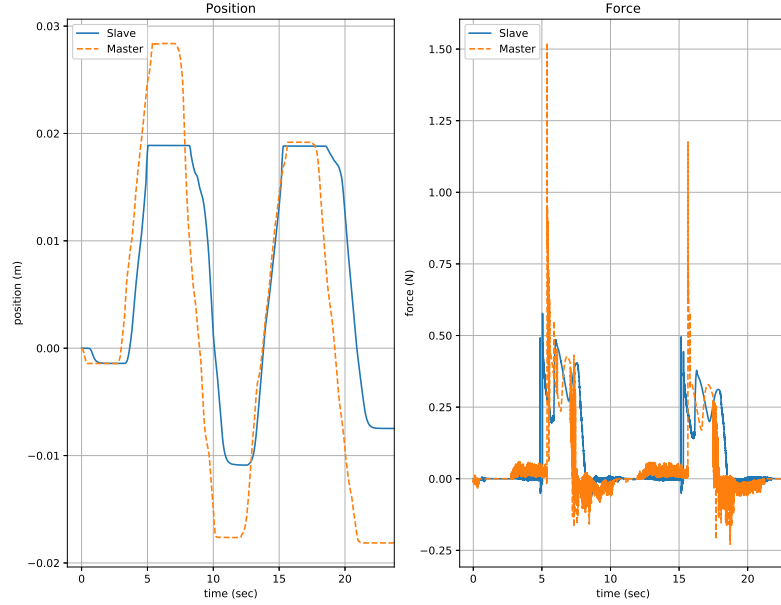


Figure 4.22: Position decoupling in experiment with the online tuning control scheme and Novint Falcon in free space with an immovable object.

with initial position drifts but is unable to keep the masters position the same when slave comes in contact with the wall. This is the cause of the lack of an online tuning subsystem, as the wave impedance is too low and the feedback forces too small to compensate for the stiffness of the operator.

All the other qualities we expected of the system like force reflection from master to slave after contact are met. But yet again we observed a strange behavior in some cases, specifically when master changes directions quickly and the slave has not "synchronised" with the position master had before the sudden change, bigger than usual resisting forces appear at the master side of the system. This behavior, we believe, is caused by the lack of the online tuning subsystem and will be resolved in the next section of our analysis, as we combine all the systems together.

Combined Model in free-space with a Wall

By Combining all three of the subsystems we get the following behavior depicted in Figure 4.23. We see that the wave-based force feedback controller is capable of handling contacts with an immovable object in its environment, transfer correctly the forces that occur, between the master and slave robot, and dissipate quickly the oscillations that happen an the event of the contact. No wave reflection are visible, the system is passive and can handle large delays in the communication, and the problems created by using only one of the subsystems are eliminated.

4.3.2 Improved Slave Model

After the previous experiments, we can say with confidence that the new Novint Falcon controller is working as intended and the teleoperation system has the correct behavior in the tasks we tested it on. For this reason we can improve the slave model, into a more complicated one, by adding the viscous and the Coulomb friction. This is a more important test for the improved slave model, because on the master side we have a real life robot with complex mechanics and not a simulation. If adding friction to the slave system had any major negative effect on the performance of the teleoperator, it is more

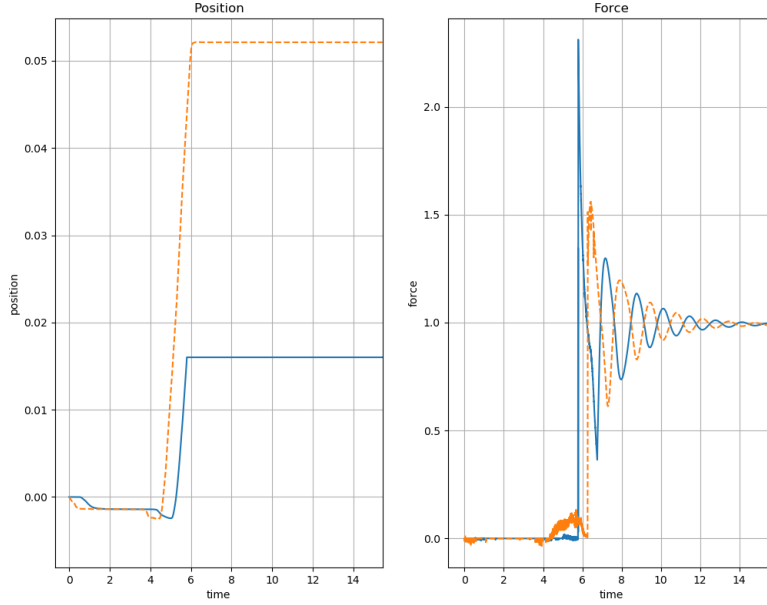


Figure 4.23: Experiment with all three subsystems combined (impedance matching, position tracking, online tuning) and Novint Falcon in free space with an immovable object.

likely to appear in the following experiments. Running the same experiments as before we start with the basic model of the teleoperator.

Basic Model in free-space

Using the basic wave matching model in freespace we get the following results. Besides the standard characteristics of this control scheme that we seen in the previous sections, here we also observe the added behavior of friction. Firstly, even though the master position is rising with a steady state, slave increases its position with a smaller rate, which is a characteristic of the Coulomb force resisting its movement by a constant amount. Then, at the start of the movement, we notice that the slave position overshoots the stable position of the master with a significant spike in slave force, something that tries to correct immediately. This is also a typical phenomenon caused by the Coulomb friction force enacting on the slave robot. The viscous friction is noticeable by the fact that the slave force is significantly higher in free movement than before, something at which the Coulomb friction also contributes.

In general the slave system appears to be stiffer even without changing the wave impedance. So far friction does not create any problems in the teleoperation, but we need to test all the subsystems one by one and then the complete system before we determine if it adds any significant problem.

Online Tuning in free-space

For this online tuning experiment we decided to increase the stiffness of the system as if we are anticipating a contact with an object, as opposed to the previous online tuning experiment without friction where we did the opposite. As in the wall environment experiments we increase the wave impedance 15cm away from the start and we try to apply the same amount of force into the haptic device.

As someone can observe, the system behaves normally until the increase in wave impedance point, where we artificially increase the stiffness of the system and the velocities involved decrease a little. The master displays larger amounts of force, distinctive

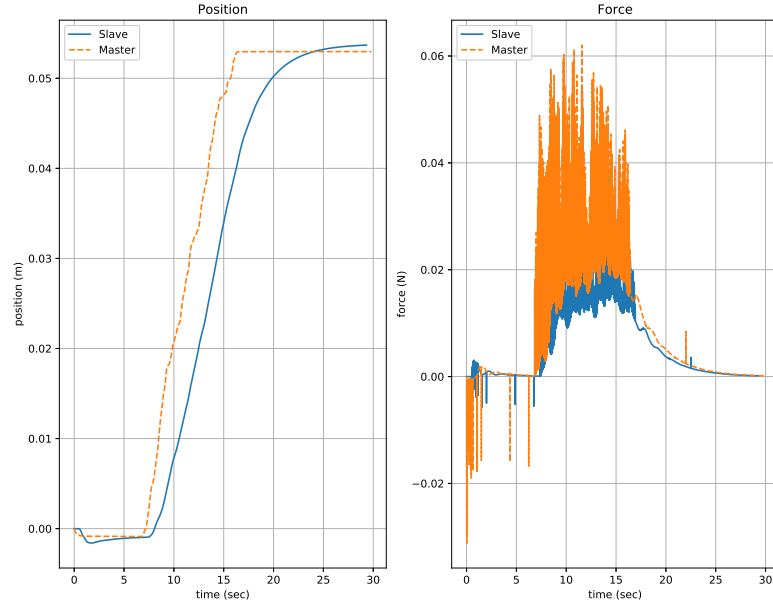


Figure 4.24: Experiment with the basic control scheme and Novint Falcon in free space with added friction.

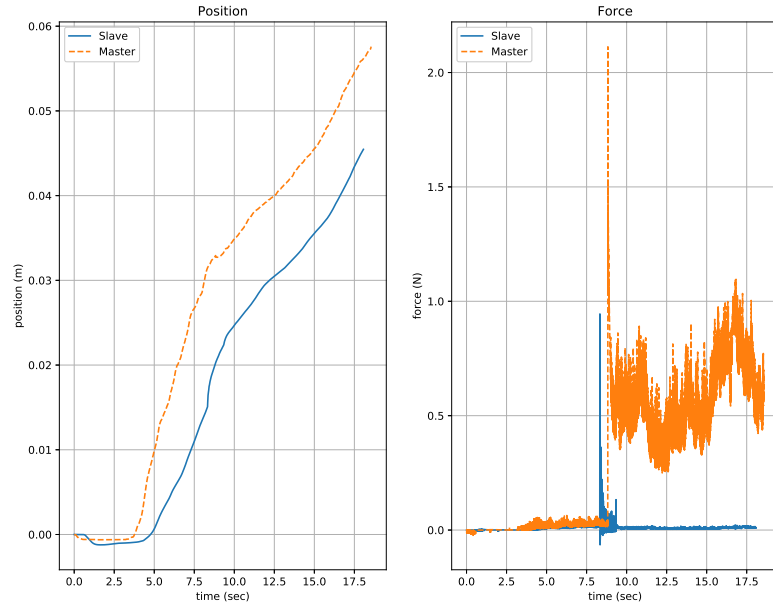


Figure 4.25: Experiment with the online tuning control scheme and Novint Falcon in free space with added friction.

of its stiffness, and a big spike where the slave impedance started to change drastically. The slave system also displays a spike at the moment of the online tuning activation, which is caused by the sharp increase of its own impedance.

This behavior is not very useful in the freespace environment as we need the system to be as maneuverable as possible and online tuning is a system required in environments with sudden changes as for example the existence of an object.

Position Tracking in free-space

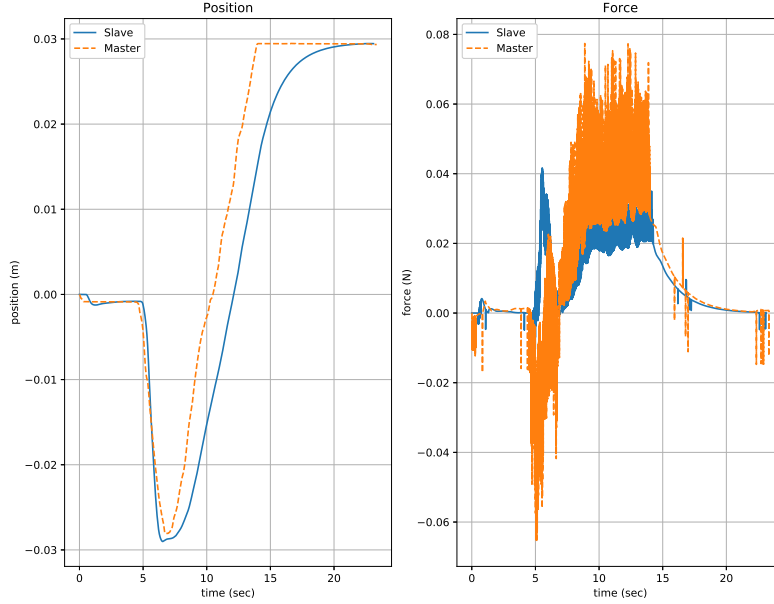


Figure 4.26: Experiment with the position tracking subsystem and Novint Falcon in free space with added friction.

Onto the position tracking model, we incorporated it onto the basic controller and we observed the same behavior as in just the basic impedance matching controller, with the exemption of a better position matching. At the start, with the friction overshoot the tow position link faster than the basic system, and at the end of the teleoperation procedure the slaves resulting position the same as of the masters. The slave and master forces act in the same way, except at the end of the first movement where the slave position surpasses that of the master so the master force inverts its sign to try and link together the two positions. And lastly the total movement is within the expected results of the controller we are analysing. Worth noticing is the fact that the slight decrease in the velocity of the slave in the steady state movement caused by the added friction is not present now. This happens because the added position tracking system prevents the two positions to drift apart.

Combined Model in free-space

Combining once again all the sub-models into one, we see a movement with the characteristics of all the subsystems as well as the existence of friction. The system is stiffer beyond the online tuning point, the two positions match each other without the slight decrease in the slaves velocity, no major wave reflections are noticeable, and finally the properties of the improved slave model are visible in the increase of the slave force in the steady state movement and the initial overshoot of the slave position.

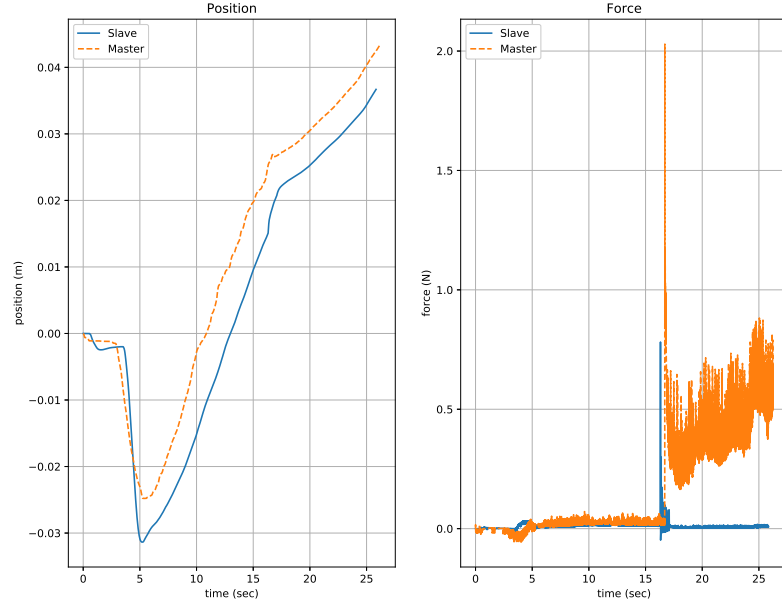


Figure 4.27: Experiment with all three subsystems combined (impedance matching, position tracking, online tuning) and Novint Falcon in free space with added friction.

Basic Model in free-space with a Wall

Testing the basic system with added friction in the case of contact with an immovable obstacle we get the following response. As we expected, the added friction increases the slave force and we find an overshoot behavior at the start of the movement as well as a slow deviation in the rate of increase of the two positions. Also the system is unable to immobilise the master robot as it lacks the online tuning behavior to stiffen up the system, but the masters force is reflected successfully onto slaves after contact with the environment is made.

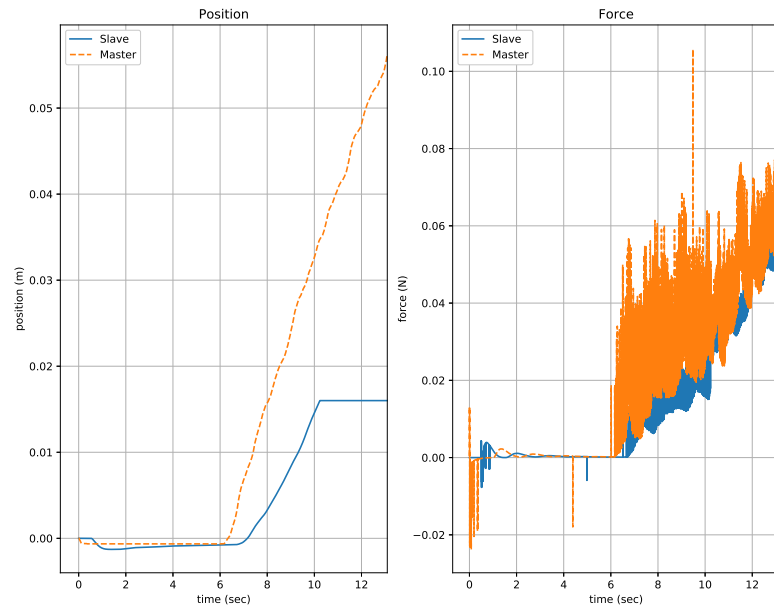


Figure 4.28: Experiment with the basic control scheme and Novint Falcon in free space with an immovable object and added friction.

Online Tuning in free-space with a Wall

On the online tuning experiment we find that the system behaves correctly after the contact with the environment. The masters movement stops after the delayed signal with the information of contact reaches, the master and slave forces are successfully reflected, while the created oscillations in these forces dissipate quickly. The characteristics that we analysed before of friction is still present but does not hinder the systems performance in any significant way.

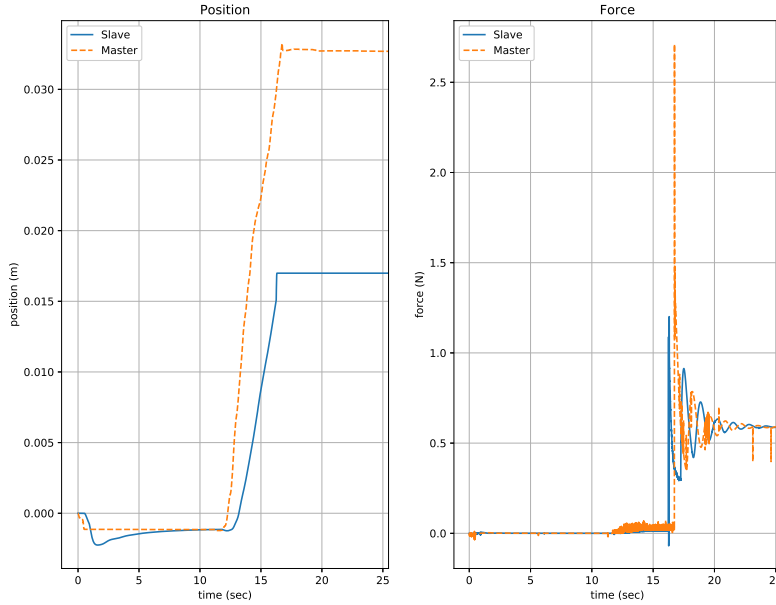


Figure 4.29: Experiment with the online tuning control scheme and Novint Falcon in free space with an immovable object and added friction.

Position Tracking in free-space with a Wall

The position tracking system does not present any new information that we did not already expect. Some characteristics of friction are negated, as the system keeps the master and slave positions coupled, relative to delay, the master force is transmitted to the environment correctly, but the environment force due to the low wave impedance is reflected significantly damped back to the operator. The result is the operator does not feel the contact with the environment and the masters movement is not obstructed.

Combined Model in free-space with a Wall

Experimenting on the combined complete teleoperation system, we find that it is fully capable of handling movement in freespace as well as a hit with an object existing in the environment. The included friction adds some characteristics to the behavior of the system but does not hinder its abilities to operate in these environments and keep the attributes of each subsystem we combined, even with large delays existing in the communication medium, while also remaining passive.

As this is our last experiment, we can say that we have explored the wave-based teleoperation controller with added impedance matching, position tracking and online tuning attributes, under one second round trip delay, and came to the conclusion that it can handle a real life task where its attributes are required.

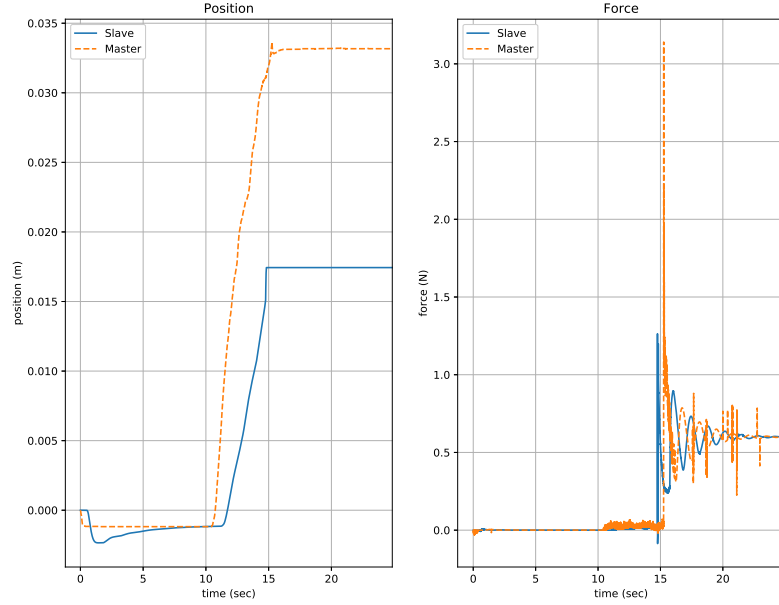


Figure 4.30: Experiment with all three subsystems combined (impedance matching, position tracking, online tuning) and Novint Falcon in free space with an immovable object and added friction.

4.3.3 Velocity Control

Thinking about the possible applications of a force feedback teleoperation control system like the one we are developing here, we get to the conclusion that matching the master and slave velocities is useful in case of a robotic arm as the slave robot, or for really short distances navigation of a rover or a flying or submerged drone for example. In cases of longer in distance movements, the control sequence is limited by the range of motion of the haptic controller.

It would be useful if we could keep the same wave-based force feedback teleoperation controller but instead of velocity to velocity matching, we match the velocity of the slave system to the position of the master system. This way we have a practically infinite workspace and for precise movements we can always revert back to the standard velocity-velocity matching controller by the press of a button. This is possible because the new capabilities will not change the system in any way that affects passivity or the way the basic wave, impedance matching and online tuning systems function, besides the addition of logic into the master controller. We will also need to remove the position tracking subsystem as its purpose is exactly opposite to the new behavior we will like to add. But it can stay in the final teleoperation scheme, stay inactive and when the operator chooses to use the old control model it can also reactivate with the rest of the old mechanics.

The velocity of the slave in this new controller, is analogous to the distance from an arbitrary home position of the master. In order to not confuse the operator a "centralising" force needs to be added, to notify the user where the home position is relative to the current position, as we as the distance from it. For this reason we added the same PID controller we used for the x and z axes onto the y axes along with the feedback force, but with lighter control constants in order to make its behavior like a soft spring. This way, the feedback forces are well transmitted from the slave to the master and vice versa, but intuitively the operator knows how much velocity and force is applied to the slave.

In practice, as we can see in the experiment we conducted with velocity control on the combined teleoperation scheme, without the position tracking subsystem for the reason we explained earlier, the system has the required behavior for a task in an environment with

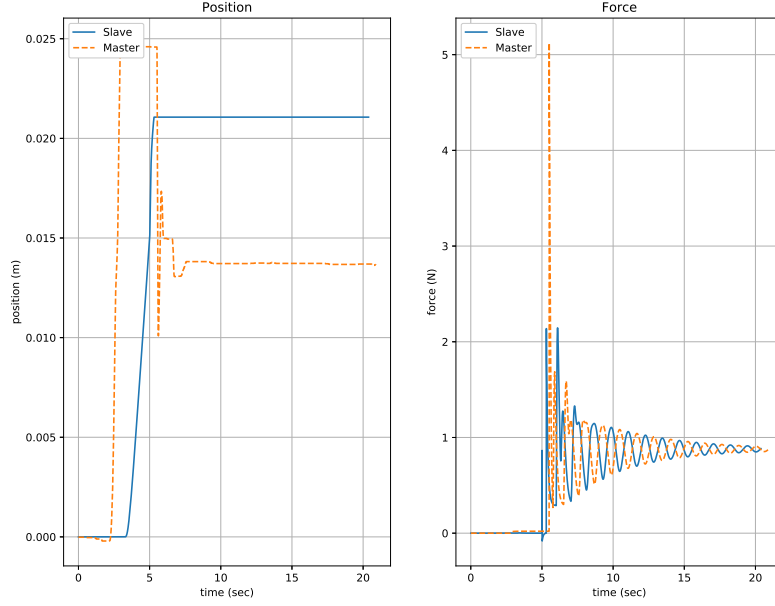


Figure 4.31: Experiment with two subsystems combined (impedance matching, online tuning) and Novint Falcon in free space with an immovable object.

an object. The forces involved are transmitted accurately to and from the master and the force oscillations dissipate after awhile. Now, we need to explain the behavior of the master system after the contact with the environment and if this behavior feels natural to the operator in order to understand the event.

When the slave robot is operating in free space, the master controller is behaving as normal, the position of the master relative to the home position is directly translated into the desired velocity of the slave and a constant force is applied to the operators hand with its direction always toward the home position (due to the extra PID controller). But when the slave robot makes contact with an object in its environment, the master force is transmitted into the environment, while the environment force transmitted and applied to the operator gives the tendency to the master to reach the home position. In practice, this translation of the master position into slave velocity, also means that the velocity of the slave is coupled with the position of the master. When the slave velocity reaches zero due to an impact with an object, the master controller applies force to the operator, trying to bring the master position back to zero. If the operator resist this tendency by applying extra force, this force will be transmitted to the environment as the slave force.

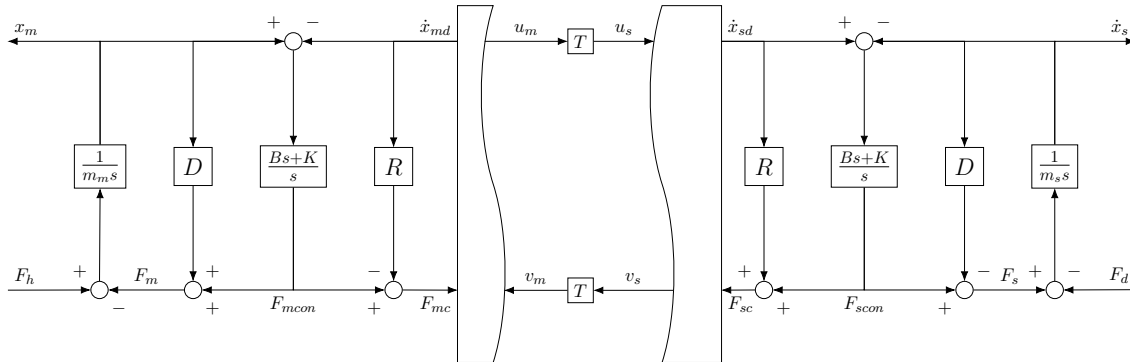


Figure 4.32: Velocity control block diagram.

We can see this behavior in Figure 4.31 where after the impact, the master position decreases, but not to zero due to the force the operator keeps applying, which in turn translates to a force applied into the object with the familiar oscillations we observed in the experiments before. If left uninterrupted, the master position will return to zero along with the slave force.

To conclude this segment, we find that this modification of the master behavior keeps all the characteristics of the basic wave-based force feedback teleoperation control scheme, as well as the attributes of the added subsystems of impedance matching and online tuning, while expanding its workspace and making it capable of longer teleoperated movements. This extra system can be easily reverted back into the normal control scheme we analysed before, while in operation, and enable finer movements.

Chapter 5

Conclusions and Future Work

In this thesis, the problem of stability in delayed teleoperation control was tackled through the method of passivity and wave variables. The wave-based approach was analysed, its means of battling instability were discussed and the inherent problems of the method were presented. Four techniques were used to counter the disadvantages of the wave variables approach, wave reflections and position drift. The Impedance Matching and Transmitting the Wave Command methods along with the Online Tuning process were further explored experimentally, in addition to their impact on one another. The experiments were conducted firstly in a simulation and then with a haptic device (Novint Falcon) as well as an improved slave model was introduced to further improve the model and get closer to a real life scenario. Finally, a new velocity controller was constructed to expand the systems workspace along with its real life applications.

In the experiments conducted, we concluded that, a wave-based teleoperation controller in cooperation with an impedance matching and a position tracking design, supplemented with the online tuning process, is fully capable of dealing with all the inherent problems of the wave variables method in tasks comprised of only simple movement in freespace, push-pull tasks where the slave robot is in contact with an obstacle, any combination of these two environments as well as advanced robot models from simulation with added friction to a real life haptic device. The passivity criteria are preserved with each complication of the basic control scheme and the delay tested was up to two seconds round trip time. Lastly, the new velocity control capabilities were added successfully to the rest of the system, making it more flexible.

Moving forward from this thesis, a complete three degrees of freedom wave-based controller can be explored and experimented on. Additionally, the slave model can be upgraded to a real life robot manipulator for the standard system and a rover or a drone for the added velocity control scheme, so true-to-real application experiments can be conducted. Even some advanced methods like "Wave prediction" or "Accelerated waves" (Sun, Naghdy, and Du 2014) can be implemented in order to further improve the performance and telepresence of a wave-based controller.

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